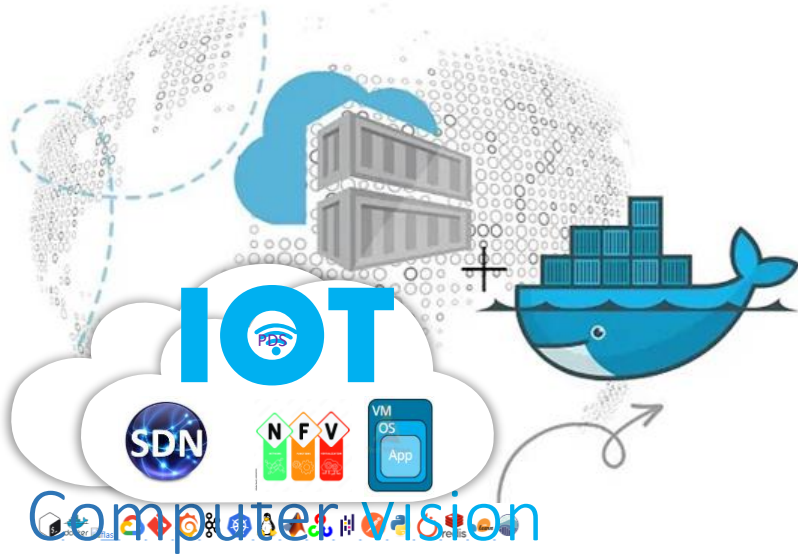
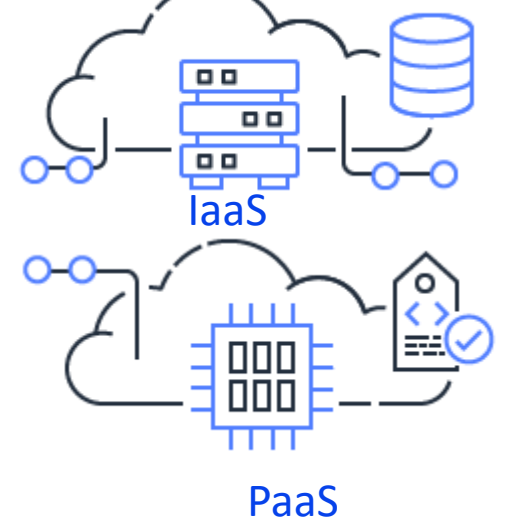
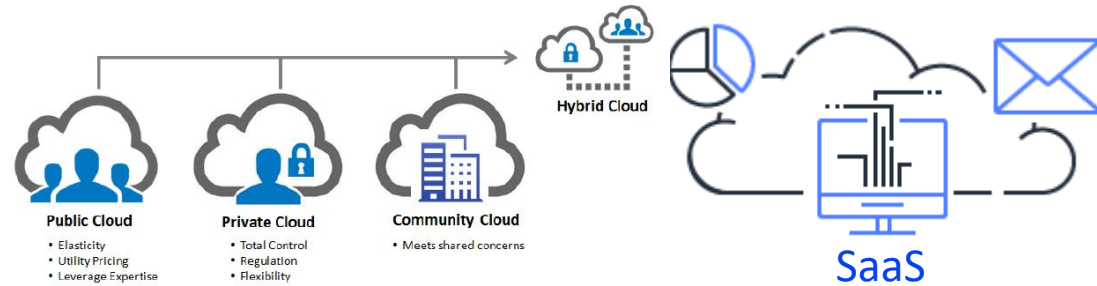
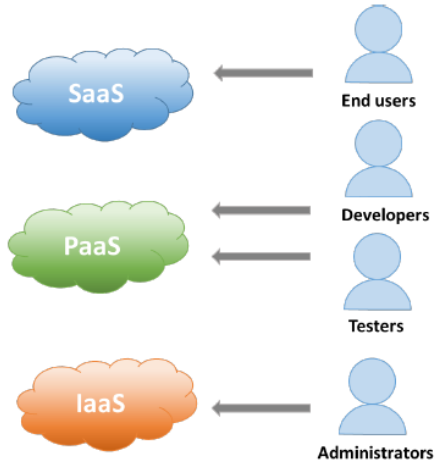




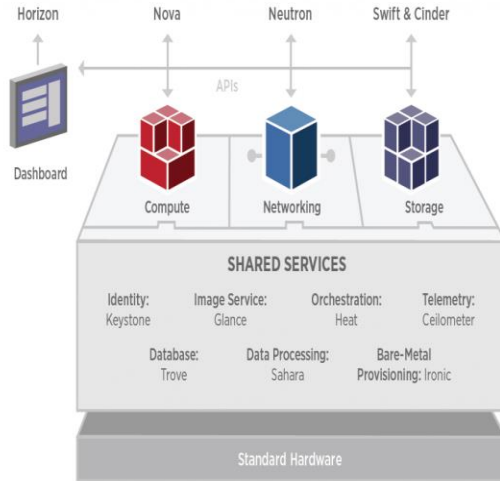
Kolla Ansible OpenStack Installation (Ubuntu server 22.04)

.

<https://github.com/siagianp>
<https://www.youtube.com/@AmelOline/videos>
https://github.com/amelcharolinesgn2/loT_simulator-mqtt-NodeRed
<https://github.com/amelcharolinesgn2/Cloud-Infrastructures>
<https://github.com/amelcharolinesgn2/ClouD-Infrastructure-SISKA-2025>



Arsitektur Openstack



OpenStack® Services



NOVA
COMPUTE LAYER



KEYSTONE
IDENTITY



CINDER
BLOCK STORAGE



SWIFT
OBJECT STORE



HORIZON
DASHBOARD/UI



HEAT
ORCHESTRATION



GLANCE
IMAGE
MANAGEMENT



NEUTRON
NETWORKING



CEILOMETER
TELEMETRY



TROVE
DBaaS



SAHARA
DATA PROCESSING



www.rackspace.com 11

Openstack Services

Kolla Ansible melakukan deploy container untuk project Openstack berikut:

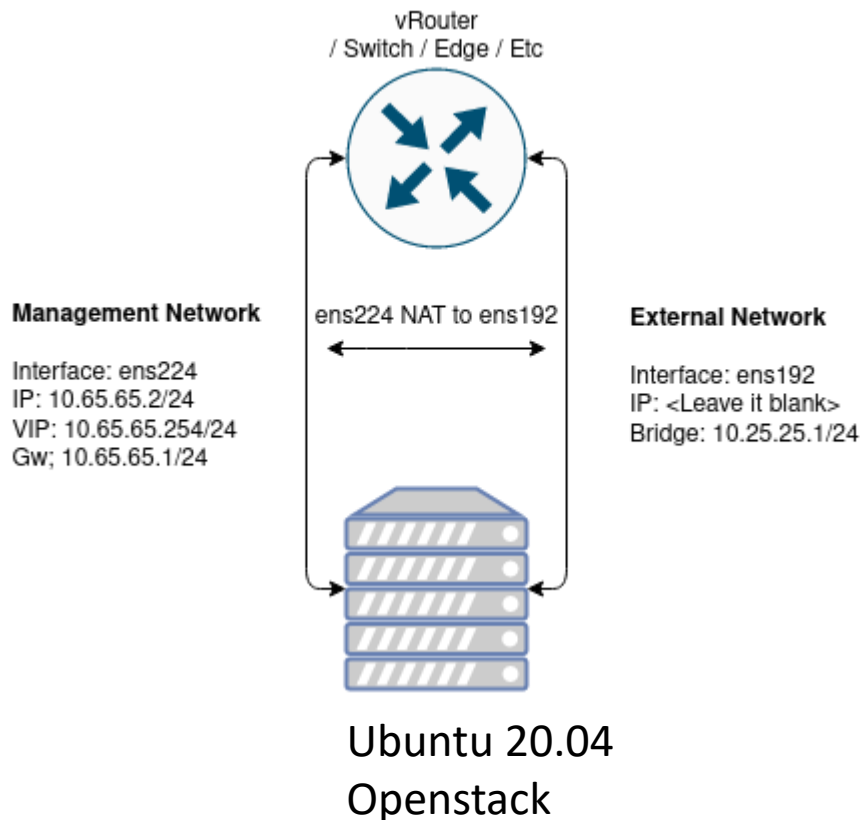
Aodh
Barbican
Bifrost
Blazar
Ceilometer
Cinder
CloudKitty
Congress
Designate
Freezer
Glance
Heat
Horizon
Ironic
Karbor
Keystone
Kuryr
Magnum
Manila
Mistral
Murano

Infrastructure components

Service Openstack berikut:

- ✓ Ceph implementation for Cinder, Glance and Nova.
- ✓ Collectd, Telegraf, InfluxDB, Prometheus, and Grafana for performance monitoring.
- ✓ Elasticsearch and Kibana to search, analyze, and visualize log messages.
- ✓ Etcd a distributed reliable key-value store.
- ✓ Fluentd as an open source data collector for unified logging layer.
- ✓ Gnocchi A time-series storage database.
- ✓ HAProxy and Keepalived for high availability of services and their endpoints.
- ✓ MariaDB and Galera Cluster for highly available MySQL databases.
- ✓ Memcached a distributed memory object caching system.
- ✓ MongoDB as a database back end for Panko.
- ✓ Open vSwitch and Linuxbridge backends for Neutron.
- ✓ RabbitMQ as a messaging backend for communication between services.
- ✓ Redis an in-memory data structure store

Lab: Kolla Ansible All in One Openstack



Requirement

Kali ini kami akan menggunakan spesifikasi mesin dan os seperti berikut, silakan dapat diikuti atau dapat disesuaikan dengan kebutuhan Anda:

- ✓ 8 CPU
- ✓ 8 GB RAM
- ✓ 100 GB Disk
- ✓ Ubuntu 20.04
- ✓ 2 Network Interface

Lab: Kolla Ansible All in One Openstack (Cons't)

2 NIC

Membutuhkan minimal 2 network interface yang nantinya akan digunakan sebagai:

1. Management Network

Interface: ens224 IP: 10.65.65.2/24 VIP: 10.65.65.254/24 Gw; 10.65.65.1/24

2. External Network

Interface: ens192 IP: kosongkan Bridge: 10.25.25.1/24

Management Network digunakan sebagai media komunikasi antar service yang menjalankan openstack, sedangkan External Network digunakan service neutron untuk berkomunikasi antara network internal (private) pada instances dan host (internet/publik), untuk interface ini diharuskan berstatus aktif (up) dan tidak dialokasikan IP pada interfacenya.

Konfigurasi Network

Saran...!!!!

Menggunakan ip statis pada setiap interface untuk menghindari perubahan IP address / routing pada saat host restart.

Network manager saya menggunakan netplan, berikut adalah konfigurasi interfacenya:

```
vi /etc/netplan/01-iface.yaml
...
# This is the network config written by 'subiquity'
network:
  ethernets:
    ens224:
      dhcp4: false
      addresses: [10.65.65.2/24]
      nameservers:
        addresses: [1.1.1.1,1.0.0.1]
    ens192: {}
  version: 2
...
netplan apply
```

Lab: Kolla Ansible All in One Openstack (Cons't)

Instalasi dependensi

Menggunakan media virtual environment yang disediakan python, adapun dependensi yang perlu diinstall adalah berikut:

```
root@administrator:/home/panda/Deploy#
```

Update paket manager:

```
# sudo apt update
```

Install dependensi python3-venv:

```
# sudo apt install python3-venv
```

Buat sebuah virtual environment lalu aktifkan untuk mulai menggunakannya:

- Buat folder baru pada direktori home **panda** dengan nama **Deploy**, bisa disesuaikan dengan fungsinya :)
- mengaktifkan virtual environment dahulu sebelum memulai menggunakan perintah python

```
mkdir deploy  
python3 -m venv /home/panda/Deploy  
source /home/panda/Deploy/bin/activate
```

Upgrade versi pip

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
root@administrator:/home/panda/Deploy#
```

```
# pip install -U pip
```

Install paket ansible :

Kolla ansible membutuhkan paket ansible minimal versi 4 dan mendukung hingga versi 5 keatas.

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# pip install 'ansible>=4,<6'
```

Lab: Kolla Ansible All in One Openstack (Cons't)

```
root@administrator:/home/panda/Deploy#
```

Install paket docker:

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# pip install docker
```

Install Kolla-ansible

Install kolla-ansible beserta dependensinya menggunakan python-pip

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# pip install  
git+https://opendev.org/openstack/kolla-ansible@master
```

Buat directory /etc/kolla

```
root@administrator:/home/panda/Deploy#
```

```
# sudo mkdir -p /etc/kolla
```

```
# sudo chown $panda:$panda /etc/kolla
```

Salin file global.yml dan password.yml ke directory /etc/kolla:

```
# cp -r  
/home/panda/deploy/share/kolla-  
ansible/etc_examples/kolla/*  
/etc/kolla
```

Salin file konfigurasi all-in-one:

```
# cp /path/to/venv/share/kolla-  
ansible/ansible/inventory/all-in-one  
/home/panda/
```

Tambahkan python interpreter pada konfigurasi all-in-one

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# source /home/panda/deploy/bin/activate
```


Lab: Kolla Ansible All in One Openstack (Cons't)

Cari lokasi binary python:

```
which python  
/home/panda/Deploy/bin/python
```

Tambahkan interpreter sesuai lokasi

```
vi all-in-one  
...  
Localhost  
ansible_python_interpreter=/home/panda/Deploy/bin/python
```

Install Ansible Galaxy requirements

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# kolla-ansible install-deps
```

Konfigurasi Awal

Generate Password

Password yang digunakan oleh kolla ansible disimpan di direktori **/etc/kolla/passwords.yml**

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# kolla-genpwd
```

Konfigurasi global.yml

```
vi /etc/kolla/global.yml  
...  
kolla_base_distro: "ubuntu"  
openstack_release: "master"  
kolla_internal_vip_address: "10.65.65.254"  
network_interface: "ens224"  
neutron_external_interface: "ens192"  
neutron_plugin_agent: "openvswitch"  
enable_neutron_provider_networks: "yes"  
nova_compute_virt_type: "kvm"
```

Simpan konfigurasi dan lanjut ke deployment !

Lab: Kolla Ansible All in One Openstack (Cons't)

Deployment

Kolla Ansible sudah menyediakan playbook list yang akan menginstal semua service yang sudah di set pada konfigurasi sebelumnya.

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

@ Lakukan bootstrapping pada host menggunakan dependensi kolla ansible:

```
# kolla-ansible -i all-in-one  
bootstrap-servers
```

@ Lakukan pre-checks konfigurasi

```
# kolla-ansible -i all-in-one prechecks
```

@ Jika dari hasil check sudah tidak ada error, selanjutnya deploy server

```
# kolla-ansible -i all-in-one deploy
```

Proses deployment akan membutuhkan waktu yang agak lama, pastikan saja kalian sudah menjalankan **screen** bila menggunakan remote SSH.

Jika deployment berhasil output akhirnya kurang lebih seperti berikut:

@ Lakukan bootstrapping pada host menggunakan dependensi kolla ansible:

...

PLAY RECAP

```
*****  
*****  
*****  
*****
```

```
localhost           : ok=366 changed=72  
unreachable=0  **failed=0**  skipped=245  
rescued=0  ignored=0
```

Lab: Kolla Ansible All in One Openstack (Cons't)

List Running OpenStack Docker Containers

Untuk mengecek service docker yang berjalan silakan menggunakan command ini:

```
# docker ps
```

OpenStack Post Deployment

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

Install Openstack Client

```
# pip install python-openstackclient python-neutronclient python-glanceclient
```

Generate AdminRC

```
# kolla-ansible post-deploy
```

Gunakan AdminRC

```
# source /etc/kolla/admin-openrc.sh
```

Jalankan script untuk inisiasi awal

Secara default kolla ansible sudah menyiapkan script inisiasi awal yang digunakan untuk create network demo didalam openstack.

```
vim /home/panda/deploy/share/kolla-ansible/init-runonce
...
# This EXT_NET_CIDR is your public network,that you want to connect to the internet via.
ENABLE_EXT_NET=${ENABLE_EXT_NET:-1}
EXT_NET_CIDR=${EXT_NET_CIDR:-'10.25.25.0/24'}
EXT_NET_RANGE=${EXT_NET_RANGE:-'start=10.25.25.4,end=10.25.25.253'}
EXT_NET_GATEWAY=${EXT_NET_GATEWAY:-'10.25.25.1'}
```

Lab: Kolla Ansible All in One Openstack (Cons't)

Eksekusi script

```
/home/panda/deploy/share/kolla-  
ansible/init-runonce
```

Aktifkan ipv4 forwarding

```
vi /etc/sysctl.conf
```

```
...
```

```
net.ipv4.ip_forward=1  
net.ipv4.ip_nonlocal_bind=1  
net.ipv6.ip_nonlocal_bind=1  
net.unix.max_dgram_qlen=128  
net.bridge.bridge-nf-call-iptables=1  
net.bridge.bridge-nf-call-ip6tables=1  
net.ipv4.neigh.default.gc_thresh1=128  
net.ipv4.neigh.default.gc_thresh2=28672  
net.ipv4.neigh.default.gc_thresh3=32768  
net.ipv6.neigh.default.gc_thresh1=128  
net.ipv6.neigh.default.gc_thresh2=28672  
net.ipv6.neigh.default.gc_thresh3=32768
```

Sysctl

```
# sysctl -p
```

Atur interface bridge

Jalankan perintah berikut ini :

```
export ENABLE_EXT_NET=${ENABLE_EXT_NET:-1}  
export EXT_NET_CIDR=${EXT_NET_CIDR:-  
'10.25.25.0/24'}  
export EXT_NET_RANGE=${EXT_NET_RANGE:-  
'start=10.25.25.3,end=10.25.25.253'}  
export EXT_NET_GATEWAY=${EXT_NET_GATEWAY:-  
'10.25.25.1'}  
  
## Akses internet ke instances  
sudo ifconfig br-ex $EXT_NET_GATEWAY netmask  
255.255.255.0 up  
sudo iptables -t nat -A POSTROUTING -s  
$EXT_NET_CIDR -o ens224 -j MASQUERADE  
sudo iptables -A FORWARD -o ens224 -i br-ex -j  
ACCEPT  
sudo iptables -A FORWARD -i ens224 -o br-ex -j  
ACCEPT
```

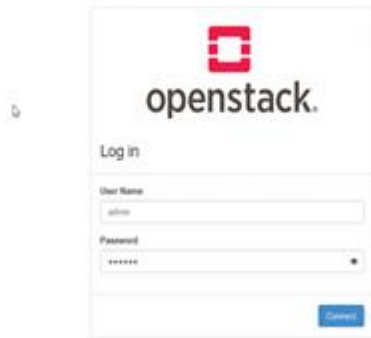
Lab: Kolla Ansible All in One Openstack (Cons't)

Akses dashboard dan uji konektifitas instance

Dashboard dapat diakses menggunakan Virtual IP yang sudah diset sebelumnya yaitu : <http://10.65.65.254>

Password dapat dilihat menggunakan command berikut ini:

```
# grep keystone_admin_password  
/etc/kolla/passwords.yml
```



Troubleshoot

Ketika host mengalami restart:

```
export EXT_NET_CIDR='10.25.25.0/24'  
export EXT_NET_RANGE='start=10.25.25.3,end=10.25.25.253'  
export EXT_NET_GATEWAY='10.25.25.1'  
sudo ifconfig br-ex $EXT_NET_GATEWAY netmask 255.255.255.0  
up  
sudo iptables -t nat -A POSTROUTING -s $EXT_NET_CIDR -o  
ens224 -j MASQUERADE  
echo 1 > /proc/sys/net/ipv4/ip_forward  
sudo iptables -A FORWARD -o ens224 -i br-ex -j ACCEPT  
sudo iptables -A FORWARD -i ens224 -o br-ex -j ACCEPT
```

Membuat Instance Melalui Horizon

- ✓ Openstack: Upload Image atau ISO
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

- ✓ **Openstack: Upload Image atau ISO**
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

✓ Upload Image atau ISO

Create Instance

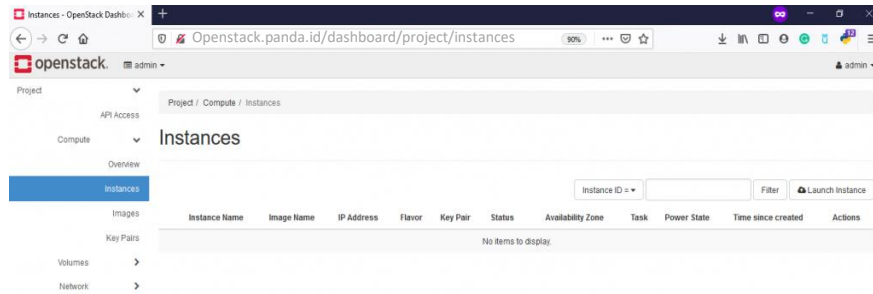
Lab: Kolla Ansible All in One Openstack (Cons't)

Upload Image atau ISO

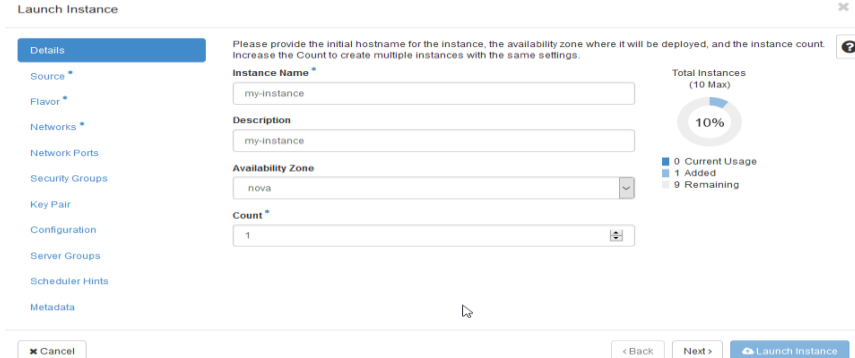
Create Instance

Upload image atau iso menggunakan cirros, silakan

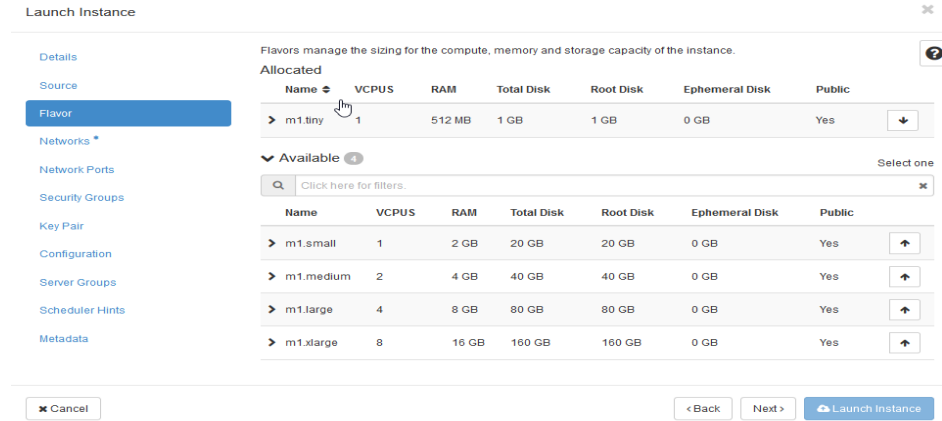
Login Horizon >> Project >> Compute >> Instances >> Launch Instance



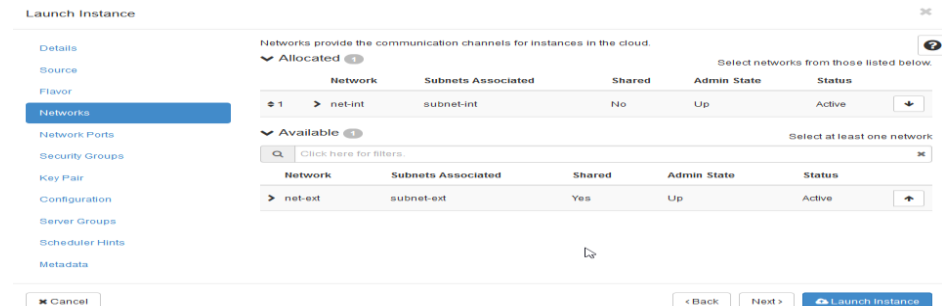
Pada menu Source silakan pilih OS atau image yang ingin Anda gunakan, disini kami contohkan OS yang kami gunakan yaitu Cirros



Menu **Flavor** silakan pilih spesifikasi instance



Menu **Networks** silakan pilih network internal yang telah dibuat sebelumnya.



Lab: Kolla Ansible All in One Openstack (Cons't)

Upload Image atau ISO

Create Instance

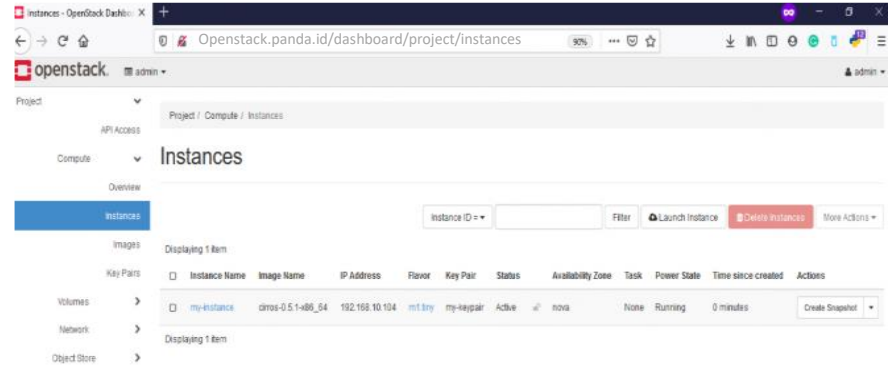
Menu Security Groups , silakan pilih Security Groups yang telah dibuat sebelumnya.

The screenshot shows the 'Launch Instance' form with the 'Security Groups' tab selected. On the left, a sidebar lists various configuration options: Details, Source, Flavor, Networks, Network Ports, Security Groups (highlighted), Key Pair, Configuration, Server Groups, Scheduler Hints, and Metadata. The main area is titled 'Select the security groups to launch the instance in.' and contains two sections: 'Allocated' and 'Available'. The 'Available' section has a search bar and a table with two columns: 'Name' and 'Description'. One security group is listed: 'default' with the description 'Default security group'. At the bottom of the form, there are buttons for 'Cancel', '< Back', 'Next >', and 'Launch Instance'.

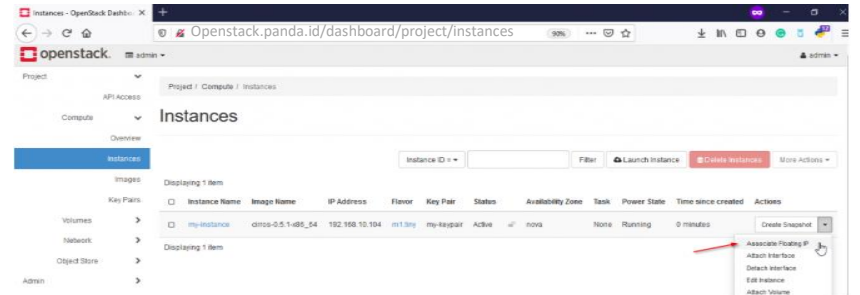
Menu **Key Pair** silakan pilih key pair yang telah di buat juga sebelumnya

The screenshot shows the 'Launch Instance' form with the 'Key Pair' tab selected. The sidebar on the left is the same as the previous screenshot, with 'Key Pair' now highlighted. The main area is titled 'A key pair allows you to SSH into your newly created instance. You may select an existing key pair, import a key pair, or generate a new key pair.' and contains buttons for 'Create Key Pair' and 'Import Key Pair'. Below this, there are sections for 'Allocated' and 'Available' key pairs. The 'Available' section has a search bar and a table with two columns: 'Name' and 'Fingerprint'. One key pair is listed: 'new-keypair' with the fingerprint '50:51:04:6a:c4:3d:21:12:13:76:8a:50:53:14:2f:93'. At the bottom, there are buttons for 'Cancel', '< Back', 'Next >', and 'Launch Instance'.

Jika sudah silakan, create **Launch Instance** silakan tunggu proses pembuatan Instance selesai



Saat ini instance sudah mendapatkan IP internal secara otomatis, supaya dapat diakses dari public silakan membuat floating IP pada instance silakan klik menu Associate Floating IP

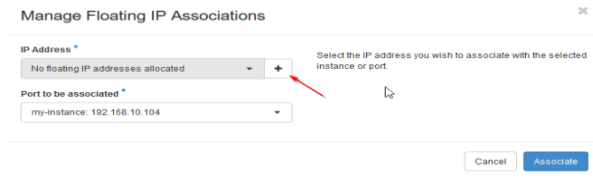


Klik tanda +

Lab: Kolla Ansible All in One Openstack (Cons't)

Upload Image atau ISO Create Instance

Manage Floating IP Associations



Manage Floating IP Associations

IP Address *
No floating IP addresses allocated

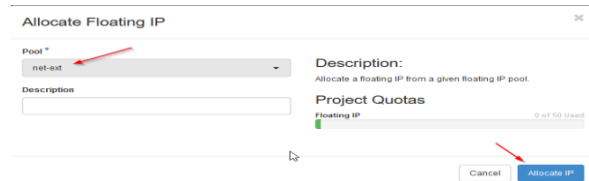
Port to be associated *
my-instance: 192.168.10.104

Select the IP address you wish to associate with the selected instance or port.

Cancel Associate

Pilih network external >> Allocate IP

Pilih network external >> Allocate IP



Allocate Floating IP

Pool *
net-ext

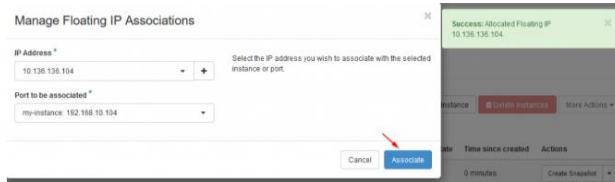
Description:
Allocate a floating IP from a given floating IP pool.

Description

Project Quotas
Floating IP 0 of 60 Used

Cancel Allocate IP

Saat ini kita sudah mendapatkan IP nya, silakan klik **Associate** untuk dapat di assign ke instance Anda



Manage Floating IP Associations

IP Address *
10.136.136.104

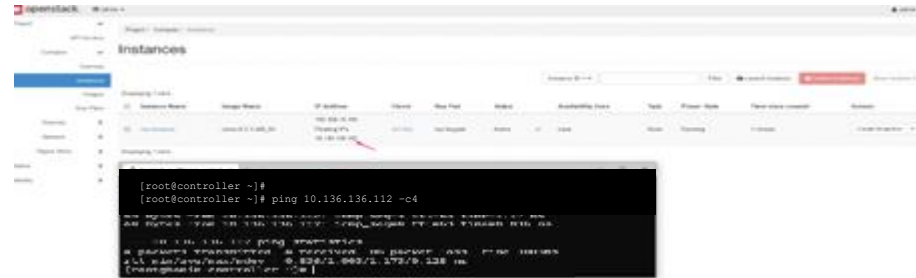
Port to be associated *
my-instance: 192.168.10.104

Select the IP address you wish to associate with the selected instance or port.

Cancel Associate

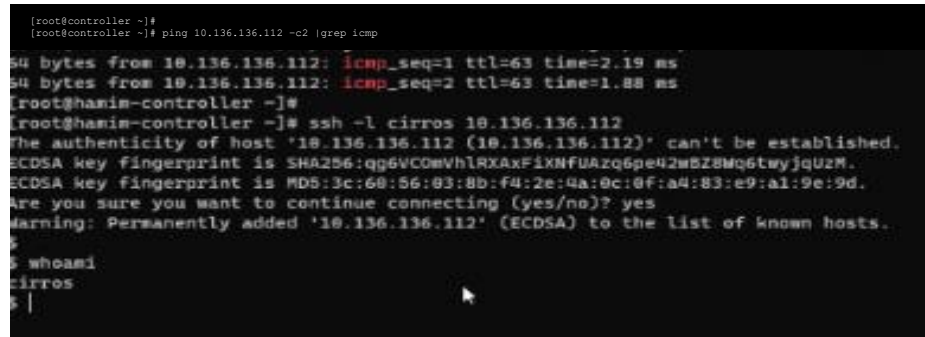
Saat ini instance sudah mendapatkan dan menggunakan IP dari network internal dan external dan silakan test ping ke IP floating

```
[root@controller ~]#  
[root@controller ~]# ping 10.136.136.112 -c4
```



Test login ke sisi instance

```
[root@controller ~]#  
[root@controller ~]# ping 10.136.136.112 -c2 |grep icmp
```



Lab: Kolla Ansible All in One Openstack (Cons't)

Upload Image atau ISO

Create Image

upload image atau iso di openstack terdapat 2 cara yang dapat dilakukan yaitu melalui horizon (melalui web) atau CLI.

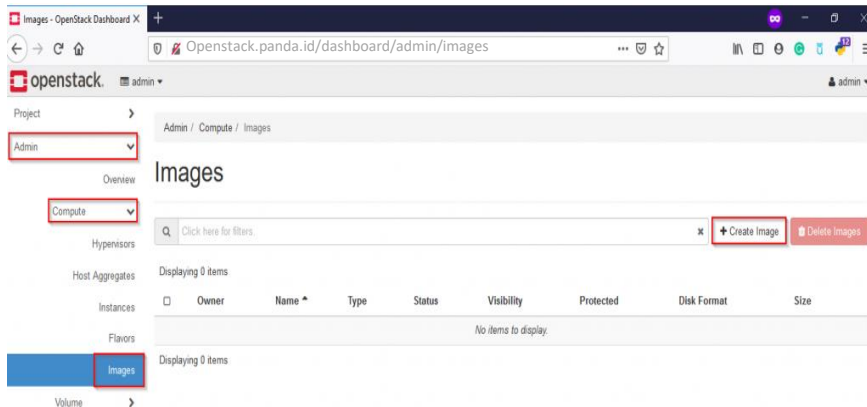
Upload image atau iso pastikan Anda sudah mengunggah image nya terlebih dahulu berikut link jika menggunakan Ubuntu, CentOS dan Cirros

[Ubuntu Cloud Images](#)

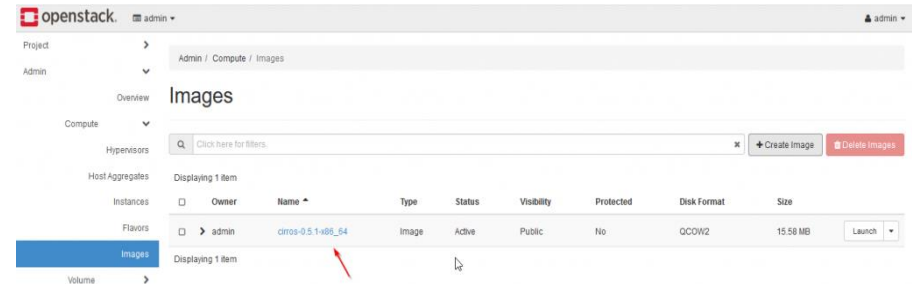
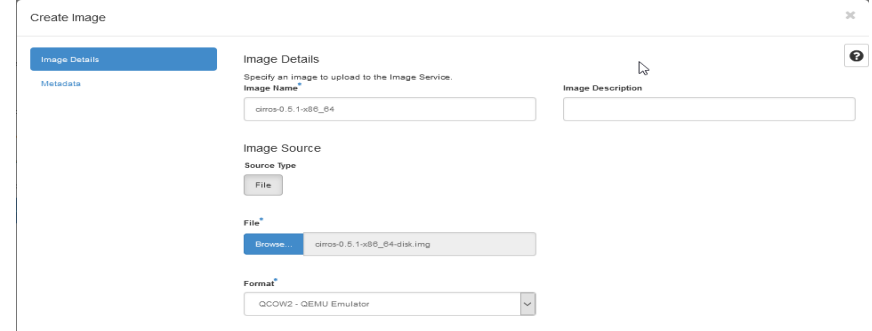
[CentOS Cloud Images](#)

[Cirros](#)

Upload image atau iso menggunakan cirros, silakan login Horizon >> Admin >> Compute >> Images >> Create Images



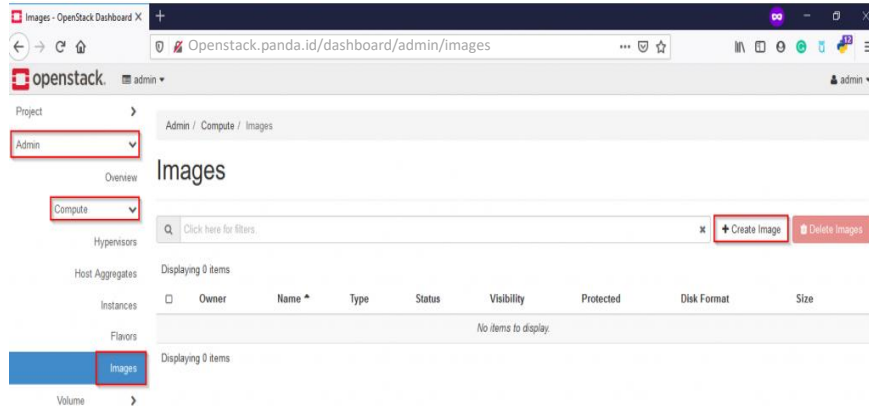
Selanjutnya, isi *Image Name* dengan nama image yang Anda inginkan dan klik *Browser >> Upload Image >> Format (QCOW2 – QEMU Emulator) >> Klik Create Image*



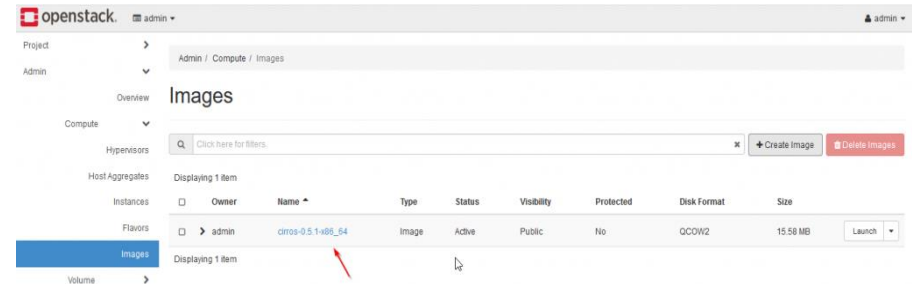
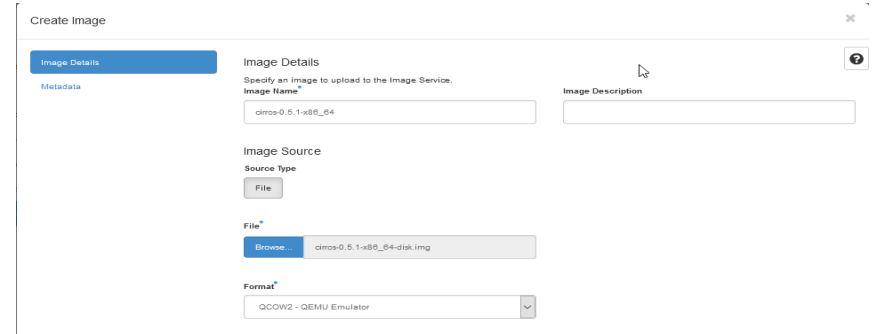
Lab: Kolla Ansible All in One Openstack (Cons't)

Create Instance Melalui Horizon

Upload image atau iso menggunakan cirros, silakan login Horizon >> Admin >> Compute >> Images >> Create Images



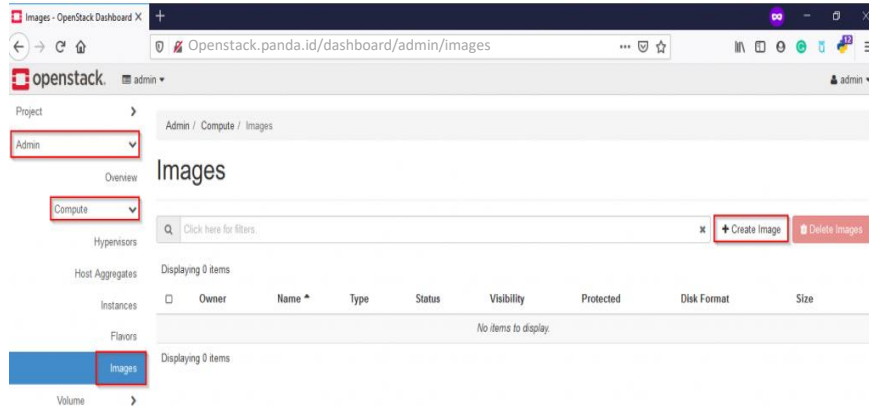
Selanjutnya, isi *Image Name* dengan nama image yang Anda inginkan dan klik *Browser* >> *Upload Image* >> *Format (QCOW2 – QEMU Emulator)* >> *Klik Create Image*



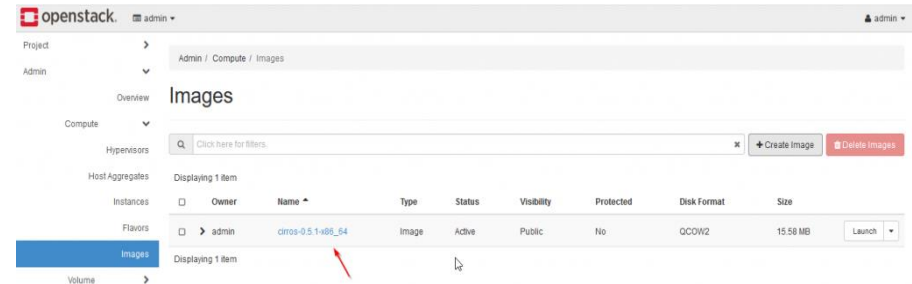
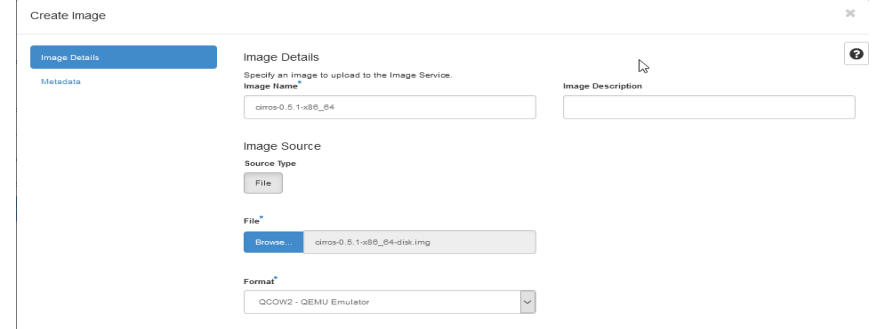
Lab: Kolla Ansible All in One Openstack (Cons't)

Create Instance Melalui Horizon

Upload image atau iso menggunakan cirros, silakan login Horizon >> Admin >> Compute >> Images >> Create Images



Selanjutnya, isi *Image Name* dengan nama image yang Anda inginkan dan klik *Browser* >> *Upload Image* >> *Format (QCOW2 – QEMU Emulator)* >> *Klik Create Image*



- ✓ Openstack: Upload Image atau ISO
- ✓ **Openstack: Membuat Network**
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

✓ Membuat Network

Create Network

Lab: Kolla Ansible All in One Openstack (Cons't)

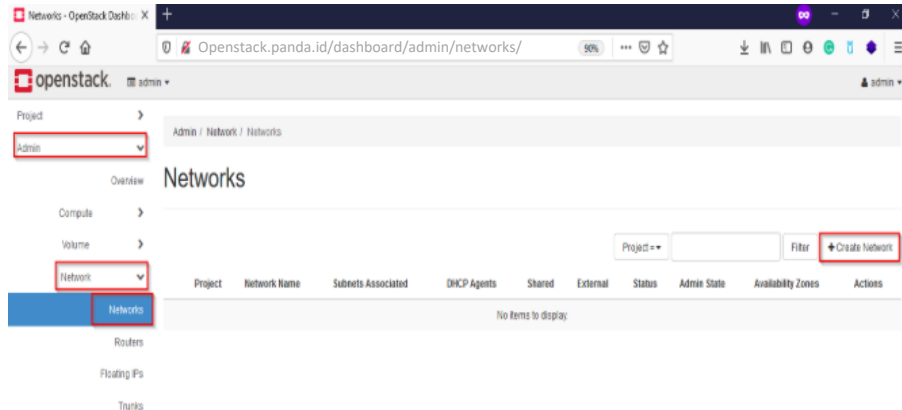
Create Network

Network yaitu network external dan network internal

- Network External: Network yang tersambung keluar (node controller dan compute).
- Network internal: Network yang akan terhubung dan digunakan oleh instance/vm.

Login Horizon >> Admin >> Network >> Networks >> Create Network

[Openstack.panda.id/dashboard/admin/networks/](https://openstack.panda.id/dashboard/admin/networks/)



Menu **Network**

A screenshot of the 'Create Network' form in the OpenStack Horizon Admin interface. The form has tabs for 'Network', 'Subnet', and 'Subnet Details', with 'Network' selected. It contains several input fields: 'Name' (with value 'net-ext'), 'Project' (dropdown with 'admin'), 'Provider Network Type' (dropdown with 'Flat'), and 'Physical Network' (with value 'extnet'). There are also checkboxes for 'Enable Admin State', 'Shared', 'External Network', and 'Create Subnet'. At the bottom, there is an 'Availability Zone Hints' field with the value 'nova'. A mouse cursor is pointing at the 'Name' field.

Name: Name network

Project: Admin

Provider Network Type: Flat

Physical Network: extnet

Selanjutnya pindah ke menu Subnet ,
silakan isi nama subnet Anda, dan isi juga
network untuk network net-ext

Lab: Kolla Ansible All in One Openstack (Cons't)

Create Network

Menu Subnet , silakan isi nama subnet Anda, dan isi juga network untuk network net-ext

Create Network

Network

Subnet

Subnet Details

Subnet Name

subnet-ext

Network Address ?

10.136.136.0/24

IP Version

IPv4

Gateway IP ?

10.136.136.1

☐ Disable Gateway

Creates a subnet associated with the network. You need to enter a valid "Network Address" and "Gateway IP". If you did not enter the "Gateway IP", the first value of a network will be assigned by default. If you do not want gateway please check the "Disable Gateway" checkbox. Advanced configuration is available by clicking on the "Subnet Details" tab.

Cancel

« Back

Next »

Menu _ Subnet Details _ silakan isi ip pool yang ingin Anda gunakan dari network address diatas, jika sudah silakan klik Create

Create Network

Network

Subnet

Subnet Details

☐ Enable DHCP

Specify additional attributes for the subnet.

Allocation Pools ?

10.136.36.100,10.136.136.199

DNS Name Servers ?

10.136.136.1

Host Routes ?

Cancel

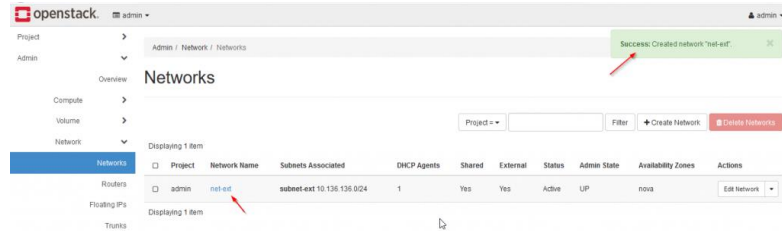
« Back

Create

Lab: Kolla Ansible All in One Openstack (Cons't)

Create Network

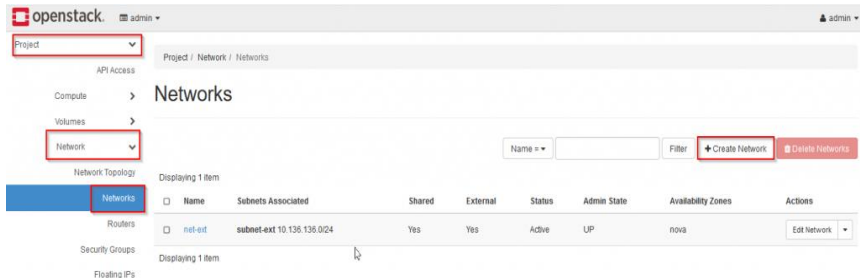
Pembuatan network selesai



Saat ini external network sudah berhasil dibuat, selanjutnya create internal network seperti berikut:

Login Horizon >> Project >> Network >> Networks >> Create Network

[Openstack.panda.id/dashboard/admin/networks/Create Network](https://openstack.panda.id/dashboard/admin/networks/CreateNetwork)



Menu Network isikan nama network

The screenshot shows the 'Create Network' wizard form. The 'Network Name' field is filled with 'net-int'. The 'Enable Admin State' checkbox is checked. The 'Create Subnet' checkbox is also checked. The 'Availability Zone Hints' dropdown is set to 'nova'.

Pindah ke menu subnet untuk menentukan ip address dan subnet nya

The screenshot shows the 'Create Subnet' wizard form. The 'Subnet Name' field is filled with 'subnet-int'. The 'Network Address' field is filled with '192.168.10.0/24'. The 'IP Version' dropdown is set to 'IPv4'. The 'Gateway IP' field is filled with '192.168.10.1'. The 'Disable Gateway' checkbox is unchecked.

Lab: Kolla Ansible All in One Openstack (Cons't)

Create Network

Pada menu Subnet Details, silakan isi pool network yang ingin digunakan dari range ip address diatas lalu klik _ Create _ untuk membuat network internal

Create Network

Network

Subnet

Subnet Details

☒ Enable DHCP

Specify additional attributes for the subnet.

Allocation Pools

192.168.10.100,192.168.10.199

DNS Name Servers

10.136.136.1

Host Routes

Cancel

« Back

Create

Proses pembuatan network internal

openstack.

admin

admin

Project

API Access

Compute

Volumes

Network

Project / Network / Networks

Success: Created network "net-int".

Networks

Name

Filter

Create Network

Delete Network

Network Topology

Displaying 2 items

	Name	Subnets Associated	Shared	External	Status	Admin State	Availability Zones	Actions
☐	net-int	subnet-int 192.168.10.0/24	No	No	Active	UP	nova	Edit Network
☐	net-ext	subnet-ext 10.136.136.0/24	Yes	Yes	Active	UP	nova	Edit Network

Displaying 2 items

- ✓ Openstack: Upload Image atau ISO
- ✓ Openstack: Membuat Network
- ✓ **Openstack: Membuat Router**
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

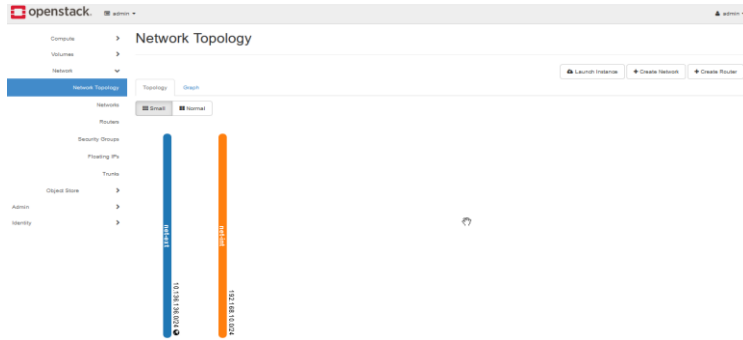
✓ Membuat Router

Create Router

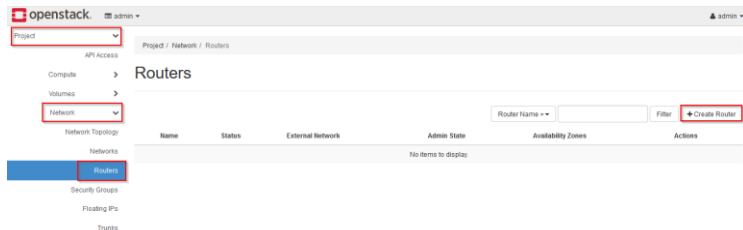
Lab: Kolla Ansible All in One Openstack (Cons't)

Create Router

Membuat router melalui horizon, fungsi router sendiri untuk menghubungkan network yang telah kita buat sebelumnya, jika di lihat melalui *Network topologi* saat ini sudah terdapat 2 network internal dan external



Membuat router Login ke Horizon >> Network >> Router >> Create Router



Isi nama Router yang Anda inginkan contoh seperti gambar dibawah ini, klik **Create Router** untuk membuat router

Create Router

Router Name: router-int

Description: Creates a router with specified parameters. Enable SNAT will only have an effect if an external network is set.

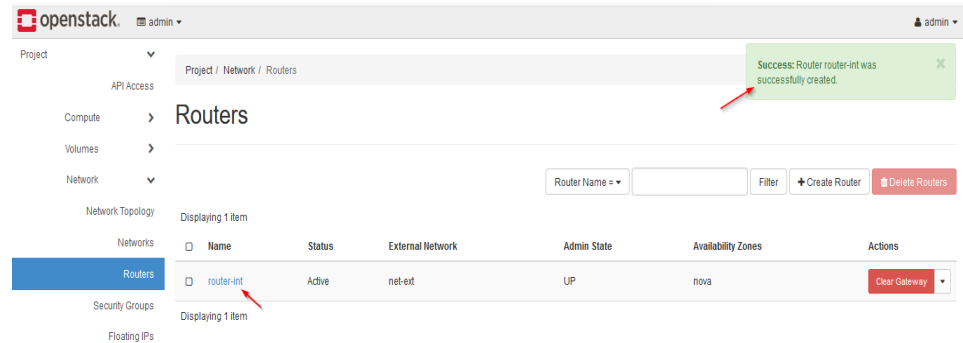
☒ Enable Admin State

External Network: net-ext

☒ Enable SNAT

Availability Zone Hints: nova

Cancel Create Router



Lab: Kolla Ansible All in One Openstack (Cons't)

Create Router

Detail informasi dari router

The screenshot shows the OpenStack dashboard with the 'router-int' router selected. The left sidebar shows the navigation menu with 'Routers' highlighted. The main content area displays the router's details:

Property	Value
Name	router-int
ID	77016449-6893-42b9-9786-23124a9ee3d
Description	
Project ID	1898e4837b2047e593534430a02992f
Status	Active
Admin State	UP
Availability Zones	nova

Below the details, the 'External Gateway' section shows the network configuration:

Property	Value
Network Name	net-int
Network ID	890a040c-5205-4201-bcda-6db8db7814fb
External Fixed IPs	Subnet ID 7258908e-146d-41c3-9668-a6e255d2599a IP Address 10.136.136.113
SNAT	Enabled

Selanjutnya pindah ke menu **Interface** >> **Add Interface** untuk menambahkan interface network yang sudah kita buat sebelumnya

The screenshot shows the OpenStack dashboard with the 'router-int' router selected. The 'Interfaces' tab is highlighted in the main content area. The 'Add Interface' button is visible in the top right corner of the interface list.

Name	Fixed IPs	Status	Type	Admin State	Actions
(#72d92f-4488)	10.136.136.113	Active	External Gateway	UP	Delete Interface

Pilih network internal yang sudah dibuat sebelumnya lalu **Submit**

The 'Add Interface' form is shown with the following fields:

- Subnet:** net-int: 192.168.10.0/24 (subnet-int)
- IP Address (optional):** (empty field)
- Description:** You can connect a specified subnet to the router. If you don't specify an IP address here, the gateway's IP address of the selected subnet will be used as the IP address of the newly created interface of the router. If the gateway's IP address is in use, you must use a different address which belongs to the selected subnet.
- Buttons:** Cancel, Submit

Saat ini interface sudah berhasil di tambahkan

The screenshot shows the OpenStack dashboard with the 'router-int' router selected. The 'Interfaces' tab is highlighted. The interface list now includes the newly added interface. A success message at the top indicates 'Success: Interface added 192.168.10.1'.

Name	Fixed IPs	Status	Type	Admin State	Actions
(2b668011-4b8d)	192.168.10.1	Down	Internal Interface	UP	Delete Interface
(#72d92f-4488)	10.136.136.113	Active	External Gateway	UP	Delete Interface

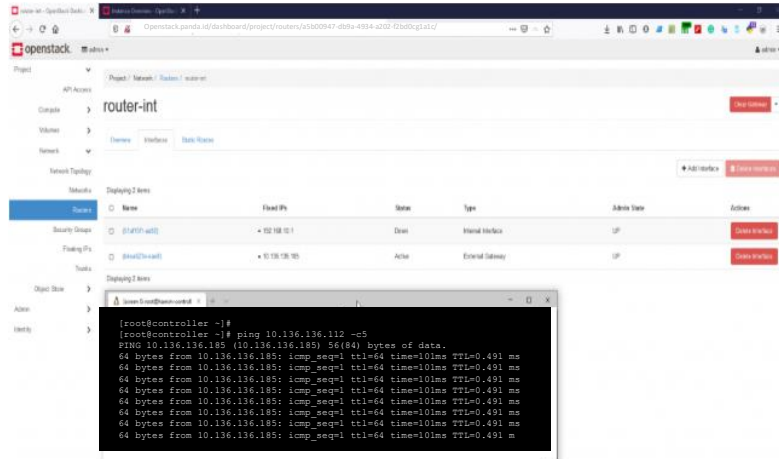
Lab: Kolla Ansible All in One Openstack (Cons't)

Create Router

Proses pembuatan network internal

Selanjutnya test ping dari controller ke IP Gateway atau IP Router yang sudah kita buat sebelumnya.

[Openstack.panda.id/dashboard/project/routers/a5b00947-db9a-4934-a202-f2bd0cg1a1c/](https://openstack.panda.id/dashboard/project/routers/a5b00947-db9a-4934-a202-f2bd0cg1a1c/)



```
[root@controller ~]#  
[root@controller ~]# ping 10.136.136.112 -c5  
PING 10.136.136.185 (10.136.136.185) 56(84) bytes of data.  
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms  
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms  
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms  
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms  
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms  
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms  
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
```

- ✓ Openstack: Upload Image atau ISO
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ **Openstack: Menambahkan SSH Key**
- ✓ Openstack: Menambah dan Menggunakan Security Group

✓ Menambahkan SSH Key

Create SSH keys

Lab: Kolla Ansible All in One Openstack (Cons't)

Create SSH Keys

Menambahkan SSH Key di openstack. Di openstack untuk menambahkan SSH Key dapat melalui CLI ataupun Horizon, disini kami contohkan menggunakan Horizon.

Login ke Horizon >> Compute >> Key Pairs
[Openstack.panda.id/dashboard/project/key_pairs](https://openstack.panda.id/dashboard/project/key_pairs)



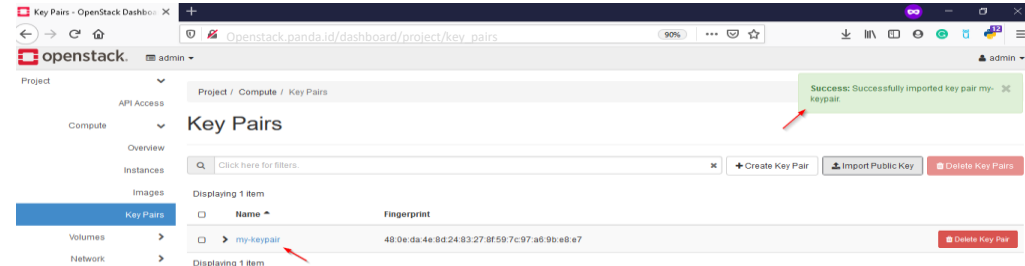
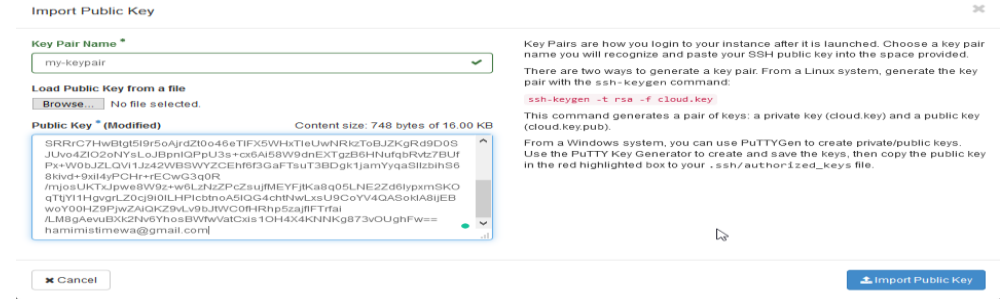
Ada 2 pemilihan terkait SSH Key diantaranya:

- Import Public Key:** Jika Anda sudah mempunyai public key pribadi Anda dapat upload public key nya tanpa create key baru.
- Create Key Pair:** Anda dapat membuat keypair baru dan keypair ini akan Anda gunakan untuk login ke instance

Berikut kami berikan contoh masing – masing pilihan diatas: Import Public Key.

✓ Import Public Key

Silakan klik menu **Import Public Key** lalu berikan nama key Anda dan isi public key Anda contoh nya seperti gambar dibawah lalu klik Import Public Key

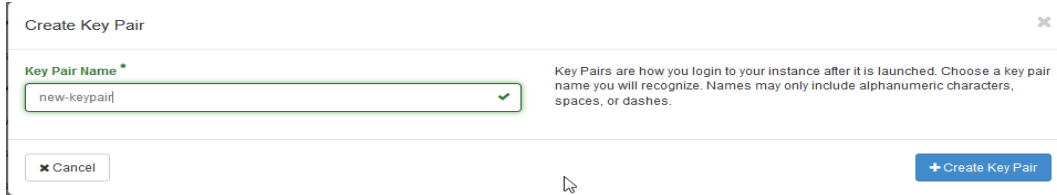


Lab: Kolla Ansible All in One Openstack (Cons't)

Create SSH Keys

Create Key Pair

Silakan Klik pada **Create Key Pair** lalu isi nama keypair



Create Key Pair

Key Pair Name *

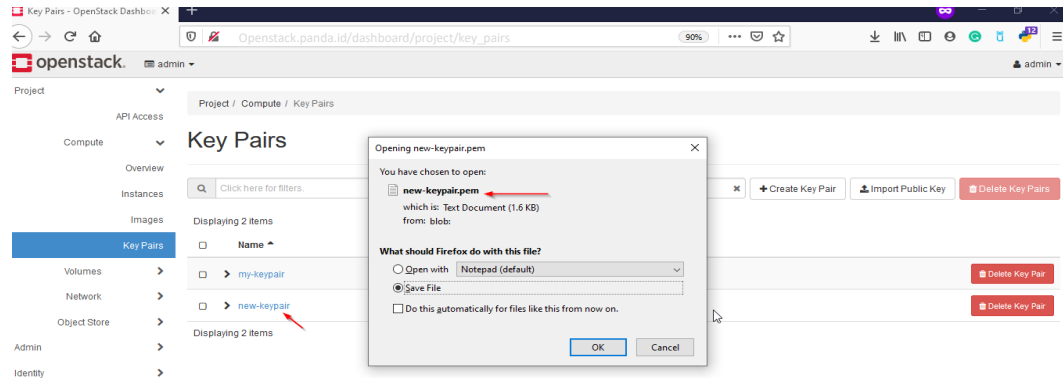
new-keypair

Key Pairs are how you login to your instance after it is launched. Choose a key pair name you will recognize. Names may only include alphanumeric characters, spaces, or dashes.

Cancel Create Key Pair

silakan klik Create Key Pair, setelah nya akan muncul pop seperti Anda, silakan download key pair nya.

[Openstack.panda.id/dashboard/project/key_pairs](https://openstack.panda.id/dashboard/project/key_pairs)



Key Pairs - OpenStack Dashboard

Openstack.panda.id/dashboard/project/key_pairs

Project / Compute / Key Pairs

Key Pairs

new-keypair.pem

You have chosen to open:

new-keypair.pem which is: Text Document (1.6 KB) from: blob

What should Firefox do with this file?

☐ Open with Notepad (default)

☒ Save File

☐ Do this automatically for files like this from now on.

OK Cancel

new-keypair

new-keypair

- ✓ Openstack: Upload Image atau ISO
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ **Openstack: Menambah dan Menggunakan Security Group**

✓ Menambah Security

Create Security Group

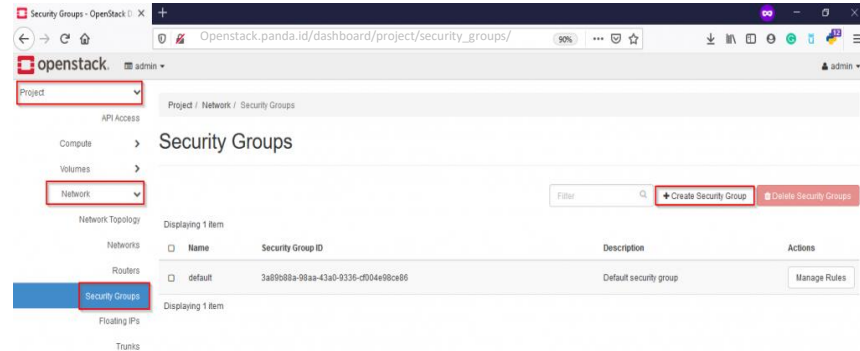
Lab: Kolla Ansible All in One Openstack (Cons't)

Create Security Group

Membuat dan menggunakan security group dapat melalui horizon atau via CLI. Disini kami akan menggunakan horizon untuk menambahkan security group, berikut tahapannya:

Login Horizon >> Project >> Network >> Security Group >> +Create Security Group

[Openstack.panda.id/dashboard/project/security_groups/](https://openstack.panda.id/dashboard/project/security_groups/)



Isi nama security group Anda lalu klik Create Security Group

The screenshot shows the 'Create Security Group' form. The 'Name' field contains 'my-secgroup'. The 'Description' field is empty. The 'Create Security Group' button is highlighted.

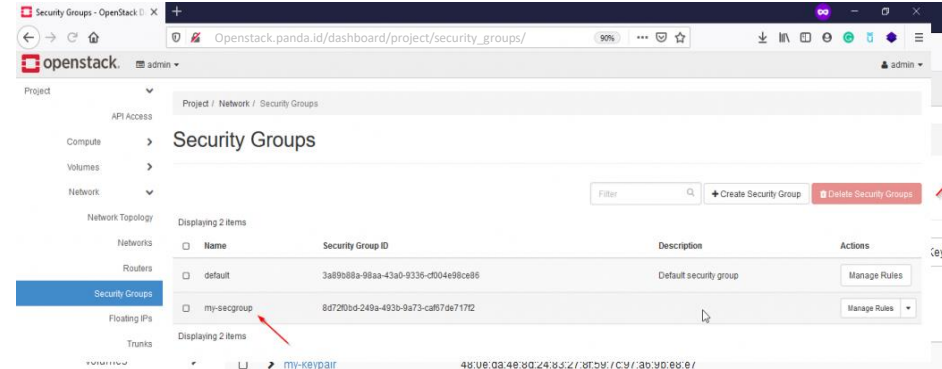
Create Security Group

Name *
my-secgroup

Description
Security groups are sets of IP filter rules that are applied to network interfaces of a VM. After the security group is created, you can add rules to the security group.

Cancel Create Security Group

Saat ini security group sudah berhasil



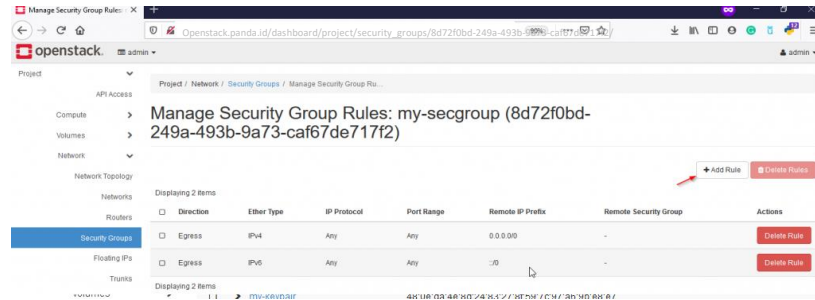
Lab: Kolla Ansible All in One Openstack (Cons't)

Create Security Group

Terdapat 2 security group (default) dan security group yang sudah dibuat sebelumnya.

Untuk yang default sebenarnya dapat Anda gunakan juga. Untuk menambahkan port atau menggunakan security group silakan klik menu **Manage Rules** pada security group Anda

[Openstack.panda.id/dashboard/project/security_groups/8d72f0bd-249a-493b-9a73-caf67de717f2/](https://openstack.panda.id/dashboard/project/security_groups/8d72f0bd-249a-493b-9a73-caf67de717f2/)



Klik + Add Rule untuk menambahkan rule atau port yang ingin Anda gunakan, misalnya disini kami ingin allow port atau protokol ICMP, lalu klik Add

Add Rule

Rule *

ALL ICMP

Direction

Ingress

Remote *

CIDR

CIDR

0.0.0.0/0

Description:

Rules define which traffic is allowed to instances assigned to the security group. A security group rule consists of three main parts:

Rule: You can specify the desired rule template or use custom rules, the options are Custom TCP Rule, Custom UDP Rule, or Custom ICMP Rule.

Open Port/Port Range: For TCP and UDP rules you may choose to open either a single port or a range of ports. Selecting the "Port Range" option will provide you with space to provide both the starting and ending ports for the range. For ICMP rules you instead specify an ICMP type and code in the spaces provided.

Remote: You must specify the source of the traffic to be allowed via this rule. You may do so either in the form of an IP address block (CIDR) or via a source group (Security Group). Selecting a security group as the source will allow any other instance in that security group access to any other instance via this rule.

Cancel

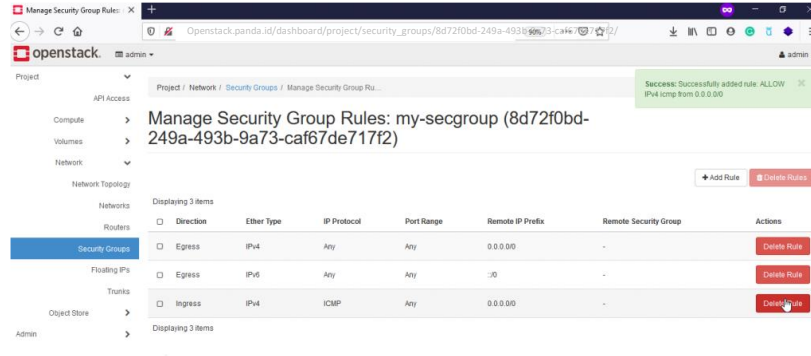
Add

Lab: Kolla Ansible All in One Openstack (Cons't)

Create SSH Keys

Saat ini rule ICMP sudah dibuat

[Openstack.panda.id/dashboard/project/security_groups/8d72f0bd-249a-493b-9a73-caf67de717f2/](https://openstack.panda.id/dashboard/project/security_groups/8d72f0bd-249a-493b-9a73-caf67de717f2/)



Dapat menambahkan rule atau port – port yang Anda inginkan misalnya port ssh, imap, http dan yang lainnya melalui security group

All Service running on Docker

Horizon



Lab: Kolla Ansible All in One Openstack (Cons't)

Network Agent

```
root@node1:~# openstack network agent list
```

ID	Agent Type	Host	Availability Zone	Alive	State	Binary
53099b35-21c1-4740-acd6-34781fbbc720	Open vSwitch agent	node1	None	:~)	UP	neutron-openvswitch-agent
6605786d-3110-43ee-b526-741b2b43abb9	L3 agent	node1	nova	:~)	UP	neutron-l3-agent
8ef397db-e6c9-418a-a532-dc6ccc330d97	DHCP agent	node1	nova	:~)	UP	neutron-dhcp-agent
b3c9da4e-d28e-4c51-b767-c4c8e4d208da	Metadata agent	node1	None	:~)	UP	neutron-metadata-agent

Compute Service

```
root@node1:~# openstack compute service list
```

ID	Binary	Host	Zone	Status	State	Updated At
058a6902-5f6c-4907-ad8e-84ec76c75f69	nova-scheduler	node1	internal	enabled	up	2024-06-15T15:57:17.000000
7c2550c7-799b-495e-ac1a-71bcab069b41	nova-conductor	node1	internal	enabled	up	2024-06-15T15:57:13.000000
b4998b8f-c07a-4b0e-9330-df2b1a9b0206	nova-compute	node1	nova	enabled	up	2024-06-15T15:57:11.000000

```
root@node1:~# █
```

Lab: Kolla Ansible All in One Openstack (Cons't)

Volume Service

```
root@node1:~# openstack volume service list
```

Binary	Host	Zone	Status	State	Updated At
cinder-scheduler	node1	nova	enabled	up	2024-06-15T15:59:53.000000
cinder-volume	node1@lvm-1	nova	enabled	up	2024-06-15T15:59:44.000000

```
root@node1:~#
```

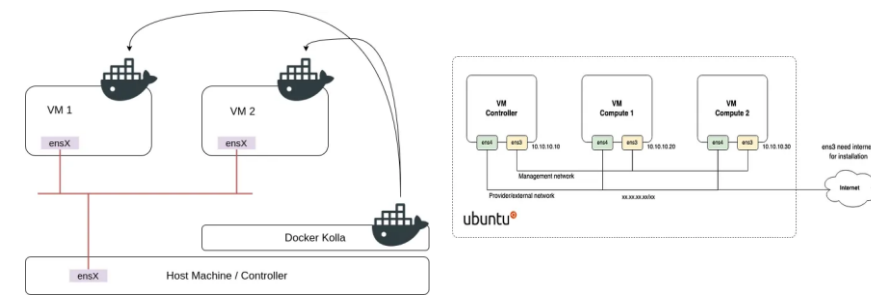
Compute Service

Deploy All-in-One OpenStack with Kolla-Ansible on Ubuntu 22.04

OpenStack Series #2 — Install openstack menggunakan Kolla Ansible (Multi-nodes)

Prasyarat

- 1. Bare-metal / Virtual Machine (Minimal 1 node untuk metode All In One dan minimal 2 node untuk metode Multinode)
- 2. Minimal 2 NIC per VM dengan 1 interface yang tidak diberikan ip
- 3. Disk tambahan untuk backend storage (At Least 50gb)
- 4. Ram minimal 8gb
- 5. Koneksi ke Internet
- 6. Mempunyai akses root maupun user sudoer



\$ openstack service list

ID	Name	Type
10cb2de52c404662aa6ca3ccdb8f2cf7	cinderv2	volumev2
250a6a7cf4d44f0c9cdd2a39f8203ccc	cinderv3	volumev3
2d99d37fc498465591faa9f44580d991	placement	placement
4001de18bf5c45f5a1467a31ea5a2afd	heat-cfn	cloudformation
505333b0b90b4962a37d55c44bc940f9	keystone	identity
612c645a37e848d49af9f6f53dc6e8ac	nova_legacy	compute_legacy
87c2edd540c84c8296251ea284a14398	neutron	network
94e54356818d43af8ed7f4e1a6da28f1	nova	compute
a9f6eb791ec740c183ecff67dc7c7b08	glance	image
eaf1f96955244acfbf5124e42c32f4e6	heat	orchestration



[Openstack.panda.id/dashboard/admin/images](https://openstack.panda.id/dashboard/admin/images)

[Openstack.panda.id/Deploy/](https://openstack.panda.id/Deploy/)

Openstack 2024.2 with Kolla Ansible

Openstack 2024.2 with Kolla Ansible (Cons't)

<i>Node Hostname</i>	<i>Node Role</i>	<i>vCPU</i>	<i>Memory</i>	<i>RootDisk</i>	<i>ManagementNet</i>	<i>StorageNet</i>
Deploy01adm01	Ansible	2 Core	2GB	20GB	10.78.78.199	-
Deploy01con01	Controller	16 Core	16GB	40GB	10.78.78.201	10.79.79.201
Deploy01con02	Controller	16 Core	16GB	40GB	10.78.78.202	10.79.79.202
Deploy01con03	Controller	16 Core	16GB	40GB	10.78.78.203	10.79.79.203
Deploy01hvp01	Hypervisor	32 Core	32GB	40GB	10.78.78.204	10.79.79.204
Deploy01hvp02	Hypervisor	32 Core	32GB	40GB	10.78.78.205	10.79.79.205



Openstack 2024.2 with Kolla Ansible (Cons't)

Interface mapping:

- ✓ eth0 for Public Network
- ✓ eth1 for Management Network
- ✓ eth2 for Storage Network with Jumbo frames
- ✓ eth4 for Tunnel/Self Service/Guest Network with Jumbo frames

We can review network configuration in **sample-netplan.yaml**



```
network:
  version: 2
  ethernet:
    eth0:
      match:
        macaddress: "BC:24:11:D3:02:2F"
      set-name: "eth0"
      critical: true
    eth1:
      match:
        macaddress: "BC:24:11:77:6F:A2"
      set-name: "eth1"
      addresses:
        - "10.78.78.201/24"
      nameservers:
        addresses:
          - 10.78.78.103
          - 10.78.78.104
        search:
          - Deploy.is-a.dev
      routes:
        - to: "default"
          via: "10.78.78.254"
      critical: true
    eth2:
      match:
        macaddress: "BC:24:11:D6:06:52"
      set-name: "eth2"
      addresses:
        - "10.79.79.201/24"
      critical: true
      mtu: 9000
    eth3:
      match:
        macaddress: "BC:24:11:EF:59:FD"
      set-name: "eth3"
      addresses:
        - "10.79.80.201/24"
      critical: true
      mtu: 9000
```

Openstack 2024.2 with Kolla Ansible (Cons't)

Installing NFS Server for Glance Service in Deploy01adm01 node

```
sudo apt update
sudo apt install nfs-kernel-server
sudo mkdir -p /data
sudo chown nobody:nogroup /data
echo "/data *(rw,sync,no_subtree_check)" >> /etc/exports
sudo systemctl restart nfs-kernel-server
sudo exportfs -v
```

Execute in all nodes

Mapping static hostname in /etc/hosts

```
cat <<EOF | tee -a /etc/hosts
10.78.78.199 Deploy01adm01 Deploy01adm01.Deploy.is-a.dev
10.78.78.201 Deploy01con01 Deploy01con01.Deploy.is-a.dev
10.78.78.202 Deploy01con02 Deploy01con02.Deploy.is-a.dev
10.78.78.203 Deploy01con02 Deploy01con03.Deploy.is-a.dev
10.78.78.204 Deploy01hvp01 Deploy01hvp01.Deploy.is-a.dev
10.78.78.205 Deploy01hvp02 Deploy01hvp02.Deploy.is-a.dev

10.78.78.200 os-int
10.78.78.100 os-ext
EOF
```

Install packages and dependencies

```
sudo apt install -y git gcc libssl-dev libffi-dev \
python3-venv python3-dev python3-selinux python3-setuptools \
python3-pip python3-docker nfs-common
```

Mount nfs volume for all node

```
sudo mkdir /data cat <<EOF | sudo tee -a /etc/fstab 10.78.78.199:/data /data nfs defaults,_netdev 0 0
EOF
sudo mount /data
```

Create virtual environment for kolla-ansible

```
python3 -m venv kolla-venv
cd kolla-venv
source bin/activate
pwd
```

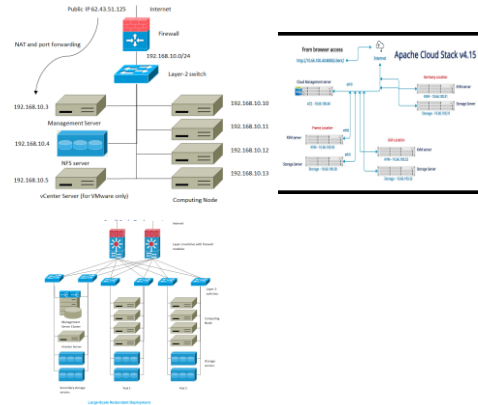
```
panda@Deploy01adm01: ~ cd kolla-venv
panda@Deploy01adm01: ~ /kolla-venv$ source bin/activate
(kolla-venv) panda@Deploy01adm01: ~ /kolla-venv $ pwd
/home/panda/ kolla-venv
(kolla-venv) panda@Deploy01adm01: ~ /kolla-venv $
```

Update pip and install ansible

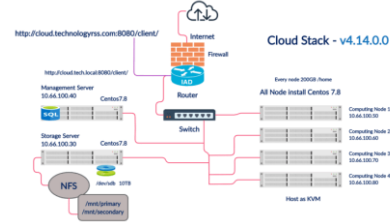
```
pip install -U pip pip install 'ansible>=8,<9'
```

All in one Multi Node cloud infrastructure (compute, Network, Storage)

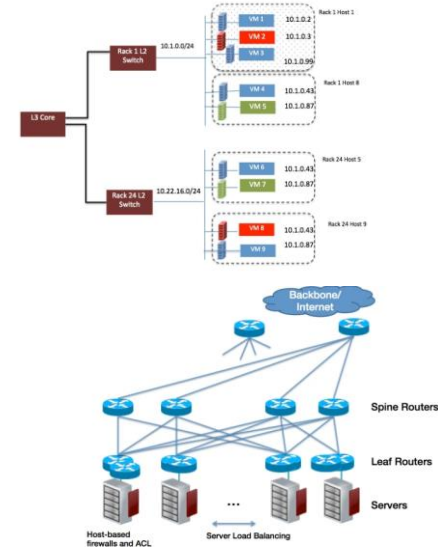
Apache Cloudstack Small-Scale Deployment



Cloud Stack



CloudStack Basic Networking



Deploy OpenStack using Kolla-Ansible (Cons't)

Lab Environment

Hosts

Hostname	Role and Spec.
GPU10	deploy (Ubuntu 22.04)
GPU111	controller (Ubuntu 20.04, 4CPU, 32G RAM, 192.168.1.23)
GPU71	compute (Ubuntu 20.04, 2CPU, 16G RAM, 192.168.1.24)
GPU72	compute (Ubuntu 20.04, 2CPU, 16G RAM, 192.168.1.25)
GPU73	compute (Ubuntu 20.04, 2CPU, 16G RAM, 192.168.1.26)

Hosts

```
eno1: internal (192.168.1.0/24)
eno2: External
```

NOTE: The eno2 interface is a "USB" NIC, rename it to eno2 base on mac address on each host.
Account username: "ubuntu"

Prerequisite

Add hosts to /etc/hosts in all hosts

```
amel@GPU10:~$ ssh-keygen
amel@GPU10:~$ ssh-copy-id GPU111
amel@GPU10:~$ ssh-copy-id GPU71
amel@GPU10:~$ ssh-copy-id GPU72
amel@GPU10:~$ ssh-copy-id GPU73
```

Configure host network interface. For example on GPU111

```
ubuntu@GPU111:~$ cat /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    eno1:
      dhcp4: false
      link-local: []
      addresses:
        - 192.168.1.23/24
      gateway4: 192.168.1.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
    eno2:
      dhcp4: false
      link-local: []
      match:
        macaddress: f8:e4:3b:87:00:1f
      set-name: eno2
  version: 2
```

```
ubuntu@GPU111:~$ sudo netplan try
ubuntu@GPU111:~$ sudo netplan apply
```

NOTE: Do not configure IP address on eno2 interface.

Add user to /etc/sudoers for no password sudo

```
ubuntu ALL=(ALL) NOPASSWD:ALL
```

NOTE: Refer "Use local docker registry to deploy" section for using local registry

Deploy OpenStack using Kolla-Ansible (Cons't)

1. Setup kolla ansible on deploy node

Install dependency packages on

```
amel@GPU10:~$ sudo apt update -y
amel@GPU10:~$ sudo apt install python3-dev libffi-dev gcc libssl-dev -y
```

Install python virtual environment for kolla-ansible

```
amel@GPU10:~$ sudo apt install python3-venv -y
```

```
amel@GPU10:~$ python3 -m venv ~/Data/kolla-ansible-venv
```

```
amel@GPU10:~$ source ~/Data/kolla-ansible-venv/bin/activate (kolla-ansible-venv) amel@GPU10:~$
```

```
(kolla-ansible-venv) amel@GPU10:~$ pip install -U pip
```

```
(kolla-ansible-venv) amel@GPU10:~$ pip install 'ansible>=4,<6'
```

Install Kolla-ansible

```
(kolla-ansible-venv) amel@GPU10:~$ pip install
git+https://opendev.org/openstack/kolla-ansible@master
```

```
(kolla-ansible-venv) amel@GPU10:~$ sudo mkdir -p /etc/
kolla(kolla-ansible-venv) amel@GPU10:~$ sudo chown $USER:$USER /etc/kolla/
(kolla-ansible-venv) amel@GPU10:~$ ls -ld /etc/kolla/
drwxr-xr-x 2 amel amel 4096 Nov 5 16:01 /etc/kolla/
```

```
(kolla-ansible-venv) amel@GPU10:~$ cp -r ~/Data/kolla-ansible-venv/share/kolla-ansible/etc_examples//kolla/* /etc/kolla/
(kolla-ansible-venv) amel@GPU10:~$ cp ~/Data/kolla-ansible-venv/share/kolla-ansible/ansible/inventory/multinode /etc/kolla/
```

NOTE: for all-in-one deploy, use ~/Data/kolla-ansible-venv/share/kolla-ansible/ansible/inventory/all-in-one

Install Ansible Galaxy requirements

```
(kolla-ansible-venv) amel@GPU10:~$ kolla-ansible install-deps
Installing Ansible Galaxy dependencies
Starting galaxy collection install process
Process install dependency map
Cloning into '/home/amel/.ansible/tmp/ansible-local-111805_n2pn0om/tmp5syatw1g/ansible-collection-kolla9pniygz2'...
remote: Enumerating objects: 150, done.
remote: Counting objects: 100% (150/150), done.
remote: Compressing objects: 100% (97/97), done.
remote: Total 409 (delta 120), reused 53 (delta 53), pack-reused 259
Receiving objects: 100% (409/409), 79.20 KiB | 477.00 KiB/s, done.
Resolving deltas: 100% (159/159), done.
Already on 'master'
Your branch is up to date with 'origin/master'.
Starting collection install process
Installing 'openstack.kolla:1.0.0' to '/home/amel/.ansible/collections/ansible_collections/openstack/kolla'
Created collection for openstack.kolla:1.0.0 at /home/amel/.ansible/collections/ansible_collections/openstack/kolla
openstack.kolla:1.0.0 was installed successfully
(kolla-ansible-venv) amel@GPU10:~$
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Configure Ansible

```
(kolla-ansible-venv) amel@GPU10:~$ cat /etc/ansible/ansible.cfg
[default]
deprecation_warnings=False
host_key_checking=False
pipelining=True
forks=100
```

Prepare initial configuration

```
(kolla-ansible-venv) amel@GPU10:~$ cd /etc/kolla/
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ ls -l
total 52
-rw-rw-r-- 1 amel amel 32632 Nov  5 16:07 globals.yml
-rw-rw-r-- 1 amel amel  9490 Nov  5 16:25 multinode
-rw-rw-r-- 1 amel amel  5163 Nov  5 16:07 passwords.yml
```

Edit multinode inventory file

```
1 # These initial groups are the only groups required to be modified. The
2 # additional groups are for more control of the environment.
3 [control]
4 # These hostname must be resolvable from your deployment host
5 GPU111
6
7 # The above can also be specified as follows:
8 #control[01:03]  ansible_user=kolla
9
10 # The network nodes are where your l3-agent and loadbalancers will run
11 # This can be the same as a host in the control group
12 [network:children]
13 control
14
15 [compute]
16 GPU7[1:3]
17
18 [monitoring:children]
19 control
20
21 # When compute nodes and control nodes use different interfaces,
22 # you need to comment out "api_interface" and other interfaces from the
23 # globals.yml
24 # and specify like below:
25 #compute01 neutron_external_interface=eth0 api_interface=em1
26 tunnel_interface=em1
27
28 [storage:children]
29 control
```

NOTE: I use "network:children", "storage:children" and "monitoring:children" to add "control" as children group. Use hostname if you do not want to use children groups.

Deploy OpenStack using Kolla-Ansible (Cons't)

Check inventory

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ ansible -i multinode all -m ping
[WARNING]: Invalid characters were found in group names but not replaced,
use -vvvv to see details
localhost | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
GPU111 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
GPU72 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
GPU73 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
GPU71 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```

Set Kolla passwords

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-genpwd
```

Edit global.yml file as below

```
48 # This should be a VIP, an unused IP on your network that will float between
49 # the hosts running keepalived for high-availability. If you want to run an
50 # All-In-One without haproxy and keepalived, you can set enable_haproxy to
no
51 # in "OpenStack options" section, and set this value to the IP of your
52 # 'network_interface' as set in the Networking section below.
53 #kolla_internal_vip_address: "10.10.10.254"
54 kolla_internal_vip_address: "192.168.1.250"

111 #####
112 # Neutron - Networking Options
113 #####
114 # This interface is what all your api services will be bound to by default.
115 # Additionally, all vxlan/tunnel and storage network traffic will go over this
116 # interface by default. This interface must contain an IP address.
117 # It is possible for hosts to have non-matching names of interfaces - these
can
118 # be set in an inventory file per host or per group or stored separately, see
119 # http://docs.ansible.com/ansible/intro\_inventory.html
120 # Yet another way to workaround the naming problem is to create a bond
for the
121 # interface on all hosts and give the bond name here. Similar strategy can be
122 # followed for other types of interfaces.
123 #network_interface: "eth0"
124 network_interface: "eno1"

149 # This is the raw interface given to neutron as its external network port. Even
150 # though an IP address can exist on this interface, it will be unusable in most
151 # configurations. It is recommended this interface not be configured with
any IP
152 # addresses for that reason.
153 #neutron_external_interface: "eth1"
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Create "nfs_shares" file with nfs configuration

```
amel@GPU10:/etc/kolla$ mkdir config
amel@GPU10:/etc/kolla$ cat config/nfs_shares
GPU111:/kolla_nfs
```

2. Setup NFS server on controller for cinder backend

```
root@GPU111:~# apt install nfs-kernel-server -y
root@GPU111:~# cat /etc/exports
/kolla_nfs 192.168.1.0/24(rw,sync,no_root_squash)
root@GPU111:~# mkdir /kolla_nfs
root@GPU111:~# systemctl start nfs-kernel-server
```

```
root@GPU111:~# systemctl status nfs-kernel-server
● nfs-server.service - NFS server and services
   Loaded: loaded (/lib/systemd/system/nfs-server.service; enabled; vendor
   preset: enabled)
   Drop-In: /run/systemd/generator/nfs-server.service.d
            └─order-with-mounts.conf
   Active: active (exited) since Sat 2025-18-05 11:51:21 UTC; 24min ago
   Process: 67275 ExecStartPre=/usr/sbin/exportfs -r (code=exited,
   status=0/SUCCESS)
   Process: 67276 ExecStart=/usr/sbin/rpc.nfsd $RPCNFSARGS (code=exited,
   status=0/SUCCESS)
   Main PID: 67276 (code=exited, status=0/SUCCESS)
```

```
Nov 05 11:51:19 GPU111 systemd[1]: Starting NFS server and services...
Nov 05 11:51:19 GPU111 exportfs[67275]: exportfs: /etc/exports [1]: Neither
'subtree_check' or 'no_subtree_check' specified for export
"192.168.1.0/24:/kolla_nfs".
Nov 05 11:51:19 GPU111 exportfs[67275]: Assuming default behaviour
('no_subtree_check').
```

3. Deployment

Bootstrap servers with kolla deploy dependencies

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-ansible -i ./multinode bootstrap-servers

... output omit

PLAY RECAP *****
localhost      : ok=2   changed=0   unreachable=0   failed=0   skipped=0   rescued=0   ignored=0
GPU111        : ok=31  changed=15  unreachable=0   failed=0   skipped=21  rescued=0   ignored=0
GPU171        : ok=30  changed=14  unreachable=0   failed=0   skipped=22  rescued=0   ignored=0
GPU172        : ok=30  changed=14  unreachable=0   failed=0   skipped=22  rescued=0   ignored=0
GPU173        : ok=30  changed=14  unreachable=0   failed=0   skipped=22  rescued=0   ignored=0
```

NOTE: if bootstrap-servers playbook failed with error. Failed to update apt cache: W:Updating from such a repository can't be done securely, and is therefore disabled by default, re-run the playbook.

Do pre-deployment checks for hosts

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-ansible -i ./multinode prechecks

... output omit

PLAY RECAP *****
localhost      : ok=11  changed=0   unreachable=0   failed=0   skipped=14  rescued=0   ignored=0
GPU111        : ok=93  changed=0   unreachable=0   failed=0   skipped=159 rescued=0   ignored=0
GPU171        : ok=29  changed=0   unreachable=0   failed=0   skipped=30  rescued=0   ignored=0
GPU172        : ok=28  changed=0   unreachable=0   failed=0   skipped=27  rescued=0   ignored=0
GPU173        : ok=28  changed=0   unreachable=0   failed=0   skipped=27  rescued=0   ignored=0
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Run actual deployment

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-ansible -i ./multinode deploy

... output omit

PLAY RECAP *****
localhost      : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
GPU111        : ok=287 changed=122 unreachable=0 failed=0 skipped=220 rescued=0 ignored=0
GPU71         : ok=59  changed=33 unreachable=0 failed=0 skipped=45  rescued=0  ignored=0
GPU72         : ok=54  changed=33 unreachable=0 failed=0 skipped=45  rescued=0  ignored=0
GPU73         : ok=54  changed=33 unreachable=0 failed=0 skipped=46  rescued=0  ignored=0
```

4. Access OpenStack

Install openstack client

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ pip install python-openstackclient -c https://releases.openstack.org/constraints/upper/master
```

Generate admin openrc and clouds.yaml

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ ls -l
total 96
-rw-r--r-- 1 amel amel 565 Nov 5 23:38 admin-openrc.sh
-rw-r--r-- 1 amel amel 573 Nov 5 23:38 clouds.yaml
drwxrwxr-x 2 amel amel 4096 Nov 5 22:54 config
-rw-rw-r-- 1 amel amel 32846 Nov 5 22:27 globals.yml
-rw-rw-r-- 1 amel amel 9475 Nov 5 16:59 multinode
-rw-rw-r-- 1 amel amel 34047 Nov 5 17:10 passwords.yml
```

Access method 1: source [admin-openrc.sh](#)

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ source admin-openrc.sh
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ openstack catalog list
```

Name	Type	Endpoints
nova	compute	RegionOne
		public: http://192.168.1.250:8774/v2.1
		internal: http://192.168.1.250:8774/v2.1
cinderv3	volume3	RegionOne
		public: http://192.168.1.250:8776/v3/496d5b465e684f20a8217fa408728ac2
		internal: http://192.168.1.250:8776/v3/496d5b465e684f20a8217fa408728ac2
placement	placement	RegionOne
		internal: http://192.168.1.250:8780
		public: http://192.168.1.250:8780
glance	image	RegionOne
		internal: http://192.168.1.250:9292
		public: http://192.168.1.250:9292
keystone	identity	RegionOne
		public: http://192.168.1.250:5000
		internal: http://192.168.1.250:5000
neutron	network	RegionOne
		public: http://192.168.1.250:9696
		internal: http://192.168.1.250:9696
heat-cfn	cloudformation	RegionOne
		internal: http://192.168.1.250:8000/v1
		public: http://192.168.1.250:8000/v1
heat	orchestration	RegionOne
		public: http://192.168.1.250:8004/v1/496d5b465e684f20a8217fa408728ac2
		internal: http://192.168.1.250:8004/v1/496d5b465e684f20a8217fa408728ac2

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ openstack compute service list
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Access method 2: use clouds.yaml

(kolla-ansible-venv) amel@GPU10:/etc/kolla\$ cat clouds.yaml

```
kolla-admin:
  auth:
    auth_url: http://192.168.1.250:5000
    project_domain_name: Default
    user_domain_name: Default
    project_name: admin
    username: admin
    password: Lo6CMreQQAwHmp3sCjR0NqagDm0fCoX7I0OBVY1c
    region_name: RegionOne
kolla-admin-internal:
  auth:
    auth_url: http://192.168.1.250:5000
    project_domain_name: Default
    user_domain_name: Default
    project_name: admin
    username: admin
    password: Lo6CMreQQAwHmp3sCjR0NqagDm0fCoX7I0OBVY1c
  interface: internal
  region_name: RegionOne
```

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cp ./clouds.yaml /etc/openstack/
OR
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ mkdir ~/.config/openstack
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cp ./clouds.yaml
~/.config/openstack
OR
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ export
OS_CLIENT_CONFIG_FILE=/etc/kolla/clouds.yaml
```

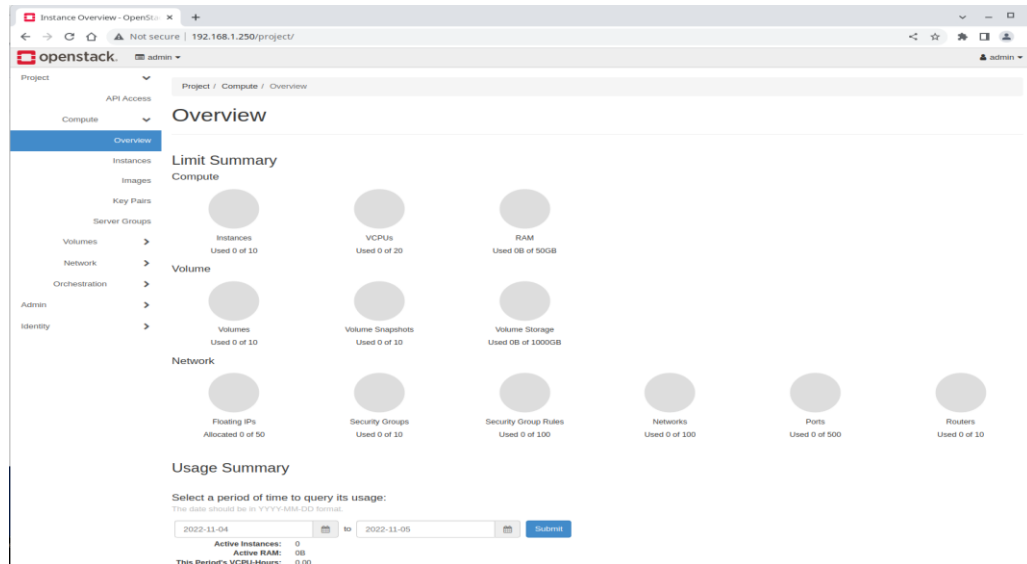
openstack command with "--os-cloud" option to choose cloud

(kolla-ansible-venv) amel@GPU10:~\$ openstack --os-cloud kolla-admin compute service list

ID	Binary	Host	Zone	Status	State	Updated At
9cb2e662-0446-417d-9ecb-d0ef5679459e	nova-scheduler	GPU111	internal	enabled	up	2025-18-05T13:05:22.000000
19dc19a4-4d78-4909-95a4-60fda0e62839	nova-conductor	GPU111	internal	enabled	up	2025-18-05T13:05:18.000000
795eefbd-9461-4462-8a68-125e77027a5b	nova-compute	GPU71	nova	enabled	up	2025-18-05T13:05:21.000000
b1734c47-918c-4aee-9fc0-78326b4dde8e	nova-compute	GPU73	nova	enabled	up	2025-18-05T13:05:21.000000
feb93682-0dab-4dc0-95cc-872d4196339d	nova-compute	GPU72	nova	enabled	up	2025-18-05T13:05:21.000000

5. Access Web GUI (Horizon)

Get admin password from admin-open.rc or clouds.yaml



Deploy OpenStack using Kolla-Ansible (Cons't)

System Information

System Information - OpenStack

Not secure | 192.168.1.250/admin/info/

openstack

admin

Project

Admin

Overview

Compute

Volume

Network

System

Defaults

Metadata Definitions

System Information

Identity

Admin / System / System Information

System Information

Services

Compute Services

Block Storage Services

Network Agents

Filter

Displaying 8 items

Name	Service	Region	Endpoints
nova	compute	RegionOne	Internal Public http://192.168.1.250:8774/v2.1 http://192.168.1.250:8774/v2.1
cinderv3	volumev3	RegionOne	Internal Public http://192.168.1.250:8776/v3/496d5b465e684f20a8217fa408728ac2 http://192.168.1.250:8776/v3/496d5b465e684f20a8217fa408728ac2
placement	placement	RegionOne	Internal Public http://192.168.1.250:8780 http://192.168.1.250:8780
glance	image	RegionOne	Internal Public http://192.168.1.250:9292 http://192.168.1.250:9292
keystone	identity	RegionOne	Internal Public http://192.168.1.250:5000 http://192.168.1.250:5000
neutron	network	RegionOne	Internal Public http://192.168.1.250:9696 http://192.168.1.250:9696
heat-cfn	cloudformation	RegionOne	Internal Public http://192.168.1.250:8000/v1 http://192.168.1.250:8000/v1
heat	orchestration	RegionOne	Internal Public http://192.168.1.250:8004/v1/496d5b465e684f20a8217fa408728ac2 http://192.168.1.250:8004/v1/496d5b465e684f20a8217fa408728ac2

Displaying 8 items

Version: 23.1.0

Deploy OpenStack using Kolla-Ansible (Cons't)

```
root@GPU111:/etc/kolla# ls -l
total 140
drwxrwx--- 2 root root 4096 Nov  5 11:44 cinder-api
drwxrwx--- 2 root root 4096 Nov  7 13:11 cinder-backup
drwxrwx--- 2 root root 4096 Nov  5 11:44 cinder-scheduler
drwxrwx--- 2 root root 4096 Nov  5 12:03 cinder-volume
drwxrwx--- 2 root root 4096 Nov  5 11:29 cron
drwxrwx--- 2 root root 4096 Nov  5 11:29 fluentd
drwxrwx--- 2 root root 4096 Nov  5 11:42 glance-api
drwxrwx--- 3 root root 4096 Nov  5 11:36 haproxy
drwxrwx--- 2 root root 4096 Nov  5 12:24 heat-api
drwxrwx--- 2 root root 4096 Nov  5 12:24 heat-api-cfn
drwxrwx--- 2 root root 4096 Nov  5 12:24 heat-engine
drwxrwx--- 2 root root 4096 Nov  5 12:25 horizon
drwxrwx--- 3 root root 4096 Nov  5 11:36 keepalived
drwxrwx--- 2 root root 4096 Nov  5 11:39 keystone
drwxrwx--- 2 root root 4096 Nov  5 11:39 keystone-fernet
drwxrwx--- 2 root root 4096 Nov  5 11:39 keystone-ssh
drwxrwx--- 2 root root 4096 Nov  5 11:29 kolla-toolbox
drwxrwx--- 2 root root 4096 Nov  5 11:37 mariadb
drwxrwx--- 2 root root 4096 Nov  5 11:37 mariadb-clustercheck
drwxrwx--- 2 root root 4096 Nov  5 11:38 memcached
drwxrwx--- 2 root root 4096 Nov  5 12:20 neutron-dhcp-agent
drwxrwx--- 2 root root 4096 Nov  5 12:21 neutron-l3-agent
drwxrwx--- 2 root root 4096 Nov  5 12:21 neutron-metadata-agent
drwxrwx--- 2 root root 4096 Nov  5 12:20 neutron-openvswitch-agent
drwxrwx--- 2 root root 4096 Nov  5 12:20 neutron-server
drwxrwx--- 2 root root 4096 Nov  5 12:07 nova-api
drwxrwx--- 2 root root 4096 Nov  5 12:05 nova-api-bootstrap
drwxrwx--- 2 root root 4096 Nov  5 12:06 nova-cell-bootstrap
drwxrwx--- 2 root root 4096 Nov  5 12:07 nova-conductor
drwxrwx--- 2 root root 4096 Nov  5 12:07 nova-novncproxy
drwxrwx--- 2 root root 4096 Nov  5 12:07 nova-scheduler
drwxrwx--- 2 root root 4096 Nov  5 12:18 openvswitch-db-server
drwxrwx--- 2 root root 4096 Nov  5 12:18 openvswitch-vswitchd
drwxrwx--- 2 root root 4096 Nov  5 12:05 placement-api
drwxrwx--- 2 root root 4096 Nov  5 11:39 rabbitmq
```

```
ubuntu@GPU71:~$ ls -al /etc/kolla/
total 44
drwxr-xr-x 11 root root 4096 Nov  5 12:20 .
drwxr-xr-x 107 root root 4096 Nov  5 12:07 ..
drwxrwx--- 2 root root 4096 Nov  5 11:29 cron
drwxrwx--- 2 root root 4096 Nov  5 11:29 fluentd
drwxrwx--- 2 root root 4096 Nov  5 11:29 kolla-toolbox
drwxrwx--- 2 root root 4096 Nov  5 12:20 neutron-openvswitch-agent
drwxrwx--- 2 root root 4096 Nov  5 12:08 nova-compute
drwxrwx--- 2 root root 4096 Nov  5 12:08 nova-libvirt
drwxrwx--- 2 root root 4096 Nov  5 12:08 nova-ssh
drwxrwx--- 2 root root 4096 Nov  5 12:18 openvswitch-db-server
drwxrwx--- 2 root root 4096 Nov  5 12:18 openvswitch-vswitchd
```

Extra: Add Ceph storage as an Cinder backend

NOTE: Refer to https://verdysgn.io/8tkrVJP7QU-MjO-0_W13jQ for setup a Ceph storage.

This section provides steps to add a Ceph storage into Cinder volume as backend; and set Nova to use the Ceph storage for running instance.

Deploy OpenStack using Kolla-Ansible (Cons't)

Setup Ceph storage via cephadm

tags: ceph

Example of Ceph storage cluster setup via cephadm.

I setup Ceph storage for OpenStack to use as "shared" storage which allow "live-migration" of instance.

OpenStack setup:
https://verdysgn.io/v2HFpS2AS8CVXiv_I2svZg (kolla-ansible)

Environment

IP Address	hostname	Ceph Node
192.168.1.23	GPU111	deploy & mon
192.168.1.24	GPU71	osd
192.168.1.25	GPU72	osd
192.168.1.26	GPU73	osd

The GPU111 will be deploy node, and initial ceph monitor host. (cephadm will auto add 2 addtional monitors automatically base on cluster size)The GPU71, GPU72 and GPU73 are OSD hosts, they have additional disk /dev/sdb (usb 3.0 drive) for OSD drive.

Steps

Create **cephadmin** user and set to sudoer

```
root@GPU111:~# useradd -m -s /bin/bash cephadmin
root@GPU111:~# passwd cephadmin
New password:
Retype new password:
passwd: password updated successfully
root@GPU111:~# echo "cephadmin ALL=(ALL) NOPASSWD:ALL" >> /etc/sudoers.d/cephadmin
root@GPU111:~# ls /etc/sudoers.d/
cephadmin  README
root@GPU111:~# chmod 0440 /etc/sudoers.d/cephadmin
```

Install cephadm tool

```
root@GPU111:~# apt install cephadm
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupdplugin1 libisns0
Use 'apt autoremove' to remove them.
Recommended packages:
  docker.io
The following NEW packages will be installed:
  cephadm
0 upgraded, 1 newly installed, 0 to remove and 11 not upgraded.
Need to get 58.5 kB of archives.
After this operation, 307 kB of additional disk space will be used.
Get:1 http://au.archive.ubuntu.com/ubuntu focal-updates/universe amd64 cephadm amd64 15.2.16-0ubuntu0.20.04.1 [58.5 kB]
Fetched 58.5 kB in 0s (665 kB/s)
Selecting previously unselected package cephadm.
(Reading database ... 130835 files and directories currently installed.)
Preparing to unpack .../cephadm_15.2.16-0ubuntu0.20.04.1_amd64.deb ...
Unpacking cephadm (15.2.16-0ubuntu0.20.04.1) ...
Setting up cephadm (15.2.16-0ubuntu0.20.04.1) ...
Adding system user cephadm....done
```

NOTE: or install via curl to fetch most recent version. For example

Deploy OpenStack using Kolla-Ansible (Cons't)

```
root@node1:~# curl --silent --remote-name --location https://github.com/ceph/ceph/raw/quincy/src/cephadm/cephadm
root@node1:~# chmod +x ./cephadm
```

Switch to **cephadmin** user

```
root@GPU111:~# su cephadmin -
cephadmin@GPU111:/root$
```

Bootstrap a new cluster

```
cephadmin@GPU111:~$ sudo cephadm bootstrap --mon-ip 192.168.1.23
Creating directory /etc/ceph for ceph.conf
Verifying podman|docker is present...
Verifying lvm2 is present...
Verifying time synchronization is in place...
Unit chrony.service is enabled and running
Repeating the final host check...
podman|docker (/usr/bin/docker) is present
systemctl is present
lvcreate is present
Unit chrony.service is enabled and running
Host looks OK
Cluster fsid: 16e21856-6030-11ed-8a24-db02e0e54b22
Verifying IP 192.168.1.23 port 3300 ...
Verifying IP 192.168.1.23 port 6789 ...
Mon IP 192.168.1.23 is in CIDR network 192.168.1.0/24
Pulling container image quay.io/ceph/ceph:v15...
Extracting ceph user uid/gid from container image...
Creating initial keys...
Creating initial monmap...
Creating mon...
Waiting for mon to start...
Waiting for mon...
mon is available
Assimilating anything we can from ceph.conf...
Generating new minimal ceph.conf
```

NOTE: add --skip-monitoring-stack option to skip monitoring stack (Prometheus, Grafana, etc...) deploy.

NOTE: add --cluster-network to use cluster network. (for example, 192.168.122.0/24) (for newer cephadm version??)

<https://github.com/ceph/ceph/pull/38911>

When login dashbaord first time, you will be asked to update password. Here I simply update the password.

```
username/password: admin/cephadmin
```

Check Ceph containers

```
cephadmin@GPU111:~$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
15225f7e7d8a	quay.io/prometheus/alertmanager:v0.20.0		3 minutes ago	Up 3 minutes		"/bin/alertmanager -..."
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-alertmanager.GPU111						
23d37d214780	quay.io/ceph/ceph-grafana:6.7.4		3 minutes ago	Up 3 minutes		"/bin/sh -c 'grafana..."
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-grafana.GPU111						
a506c9016de5	quay.io/prometheus/prometheus:v2.18.1		4 minutes ago	Up 4 minutes		"/bin/prometheus --c..."
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-prometheus.GPU111						
795ec188ff2a	quay.io/prometheus/node-exporter:v0.18.1		4 minutes ago	Up 4 minutes		"/bin/node_exporter ..."
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-node-exporter.GPU111						
a497030e7c57	quay.io/ceph/ceph:v15		5 minutes ago	Up 5 minutes		"/usr/bin/ceph-crash..."
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-crash.GPU111						
19e955a4dad6	quay.io/ceph/ceph:v15		6 minutes ago	Up 6 minutes		"/usr/bin/ceph-mgr -..."
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-mgr.GPU111.oezajx						
6120dd3bf195	quay.io/ceph/ceph:v15		6 minutes ago	Up 6 minutes		"/usr/bin/ceph-mon -..."
ceph-16e21856-6030-11ed-8a24-db02e0e54b22-mon.GPU111						

Deploy OpenStack using Kolla-Ansible (Cons't)

Check Ceph status

```
cephadmin@GPU111:~$ sudo /usr/sbin/cephadm shell --fsid 16e21856-6030-11ed-8a24-db02e0e54b22 -c /etc/ceph/ceph.conf -k /etc/ceph/ceph.client.admin.keyring
Using recent ceph image quay.io/ceph/ceph@sha256:c08064dde4bba4e72a1f55d90ca32df9ef5aafab82efe2e0a0722444a5aaacca
```

```
root@GPU111:/# sudo ceph -s
cluster:
id:   16e21856-6030-11ed-8a24-db02e0e54b22
health: HEALTH_WARN
      OSD count 0 < osd_pool_default_size 3
```

```
services:
mon: 1 daemons, quorum GPU111 (age 8m)
mgr: GPU111.oezajx(active, since 7m)
osd: 0 osds: 0 up, 0 in
```

```
data:
pools:   0 pools, 0 pgs
objects: 0 objects, 0 B
usage:   0 B used, 0 B / 0 B avail
pgs:
```

```
cephadmin@GPU111:~$ sudo ceph -s
cluster:
id:   16e21856-6030-11ed-8a24-db02e0e54b22
health: HEALTH_WARN
      OSD count 0 < osd_pool_default_size 3
```

```
services:
mon: 1 daemons, quorum GPU111 (age 12m)
mgr: GPU111.oezajx(active, since 12m)
osd: 0 osds: 0 up, 0 in
```

```
data:
pools:   0 pools, 0 pgs
objects: 0 objects, 0 B
usage:   0 B used, 0 B / 0 B avail
pgs:
```

Install Ceph-common package so we can use ceph command to check cluster directly

```
cephadmin@GPU111:~$ sudo cephadm version
Using recent ceph image quay.io/ceph/ceph@sha256:c08064dde4bba4e72a1f55d90ca32df9ef5aafab82efe2e0a0722444a5aaacca
ceph version 15.2.17 (8a82819d84cf884bd39c17e3236e0632ac146dc4) octopus (stable)
```

```
cephadmin@GPU111:~$ sudo cephadm add-repo --release octopus
Installing repo GPG key from https://download.ceph.com/keys/release.asc...
Installing repo file at /etc/apt/sources.list.d/ceph.list...
```

```
cephadmin@GPU111:~$ sudo cephadm install ceph-common
Installing packages ['ceph-common']...
```

Add host label to GPU111

```
cephadmin@GPU111:~$ sudo ceph orch host ls
HOST  ADDR  LABELS  STATUS
GPU111 GPU111
```

```
cephadmin@GPU111:~$ sudo ceph orch host label add GPU111 _admin
Added label _admin to host GPU111
```

```
cephadmin@GPU111:~$ sudo ceph orch host ls
HOST  ADDR  LABELS  STATUS
GPU111 GPU111 _admin
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Copy public key of Ceph to OSD hosts

```
cephadmin@GPU111:~$ ssh-copy-id -f -i /etc/ceph/ceph.pub root@GPU71
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/etc/ceph/ceph.pub"
The authenticity of host 'GPU71 (192.168.1.24)' can't be established.
ECDSA key fingerprint is SHA256:jwfxrwrwnLwLbIfkfJC3Yr894+FGQZnY8+yLYTNXR8.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
root@GPU71's password:
```

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'root@GPU71'"
and check to make sure that only the key(s) you wanted were added.

```
cephadmin@GPU111:~$ ssh-copy-id -f -i /etc/ceph/ceph.pub root@GPU72
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/etc/ceph/ceph.pub"
The authenticity of host 'GPU72 (192.168.1.25)' can't be established.
ECDSA key fingerprint is SHA256:9gXSkvu9mZsfNu56cQgMcSYnfEr33h5haoYs8CVxh6E.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
root@GPU72's password:
```

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'root@GPU72'"
and check to make sure that only the key(s) you wanted were added.

```
cephadmin@GPU111:~$ ssh-copy-id -f -i /etc/ceph/ceph.pub root@GPU73
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/etc/ceph/ceph.pub"
The authenticity of host 'GPU73 (192.168.1.26)' can't be established.
ECDSA key fingerprint is SHA256:4GMMsd+hO9Zh71Zrl/cHneFoPO3kBuV/f5K7JeMVhY.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
root@GPU73's password:
```

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'root@GPU73'"
and check to make sure that only the key(s) you wanted were added.

Add OSD host to cluster and set label

```
cephadmin@GPU111:~$ sudo ceph orch host add GPU71
Added host 'GPU71'
cephadmin@GPU111:~$ sudo ceph orch host add GPU72
Added host 'GPU72'
cephadmin@GPU111:~$ sudo ceph orch host add GPU73
Added host 'GPU73'
```

```
cephadmin@GPU111:~$ sudo ceph orch host ls
HOST  ADDR  LABELS  STATUS
GPU111 GPU111 _admin mon
GPU71  GPU71
GPU72  GPU72
GPU73  GPU73
```

```
cephadmin@GPU111:~$ sudo ceph orch host label add GPU71 osd
Added label osd to host GPU71
cephadmin@GPU111:~$ sudo ceph orch host label add GPU72 osd
Added label osd to host GPU72
cephadmin@GPU111:~$ sudo ceph orch host label add GPU73 osd
Added label osd to host GPU73
```

```
cephadmin@GPU111:~$ sudo ceph orch host ls
HOST  ADDR  LABELS  STATUS
GPU111 GPU111 _admin mon
GPU71  GPU71  osd
GPU72  GPU72  osd
GPU73  GPU73  osd
```

NOTE: you could tell cephadm to deploy daemons base on label. For example, to limit monitors run on host with "mon" label.

Deploy OpenStack using Kolla-Ansible (Cons't)

```
$ sudo ceph orch apply mon --placement="label:mon"
```

Add storage to cluster

```
cephadmin@GPU111:~$ sudo ceph orch
daemon add osd GPU71:/dev/sdb
Created osd(s) 0 on host 'GPU71'
cephadmin@GPU111:~$ sudo ceph orch
daemon add osd GPU72:/dev/sdb
Created osd(s) 1 on host 'GPU72'
cephadmin@GPU111:~$ sudo ceph orch
daemon add osd GPU73:/dev/sdb
Created osd(s) 2 on host 'GPU73'
```

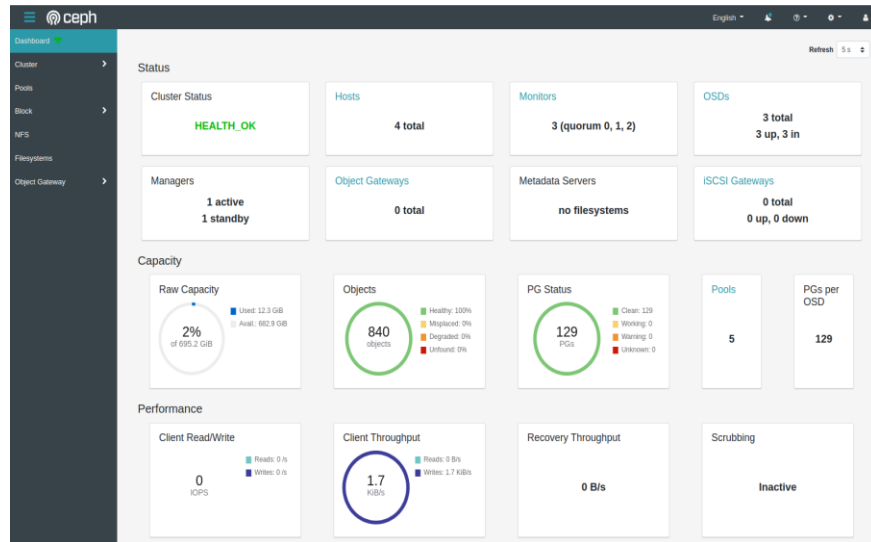
Check cluster status

```
cephadmin@GPU111:~$ sudo ceph -s
cluster:
  id:   16e21856-6030-11ed-8a24-
db02e0e54b22
  health: HEALTH_OK

services:
  mon: 3 daemons, quorum
GPU111,GPU71,GPU72 (age 119s)
  mgr: GPU111.oezajx(active, since 36m),
standbys: GPU71.pmlfuu
  osd: 3 osds: 3 up (since 24s), 3 in (since
24s)

data:
  pools:   1 pools, 1 pgs
  objects: 0 objects, 0 B
  usage:   3.0 GiB used, 692 GiB / 695 GiB
avail
  pgs:    1 active+clean
```

Dashboard



Deploy OpenStack using Kolla-Ansible (Cons't)

Purging cluster

```
root@GPU111:~# ceph mgr module disable cephadm
```

```
root@GPU111:~# ceph fsid  
16e21856-6030-11ed-8a24-db02e0e54b22
```

```
cephadm rm-cluster --force --zap-osds --fsid 16e21856-6030-11ed-8a24-db02e0e54b22
```

To disable/remove the Prometheus related services (TBC)

```
$ ceph mgr module disable prometheus  
$ ceph orch rm node-exporter
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Extra: create examples for demo

NOTE: copy "clouds.yaml" in controller node to /etc/kolla/ first.

Create pool for volume

```
root@GPU111:~# ceph osd pool create volumes
pool 'volumes' created
root@GPU111:~# ceph osd pool create vms
pool 'vms' created
root@GPU111:~# ceph osd pool create backups
pool 'backups' created
```

```
root@GPU111:~# rbd pool init volumes
root@GPU111:~# rbd pool init vms
root@GPU111:~# rbd pool init backups
```

NOTE: we still create "backups" pool because "Kolla-Ansible" playbook still add Ceph keyring for cinder-backup no matter "nfs" is configured to cinder_backup_driver. I think its kinda of bug. Fail message if no keyring for "cinder-backup" configured.

```
TASK [cinder : Copy over Ceph keyring files for cinder-backup]
*****
```

```
An exception occurred during task execution. To see the full traceback, use -vvv. The error was: If you are using a
module and expect the file to exist on the remote, see the remote_src option
failed: [GPU111] (item=cinder-backup/ceph.client.cinder.keyring) => {"ansible_loop_var": "item", "changed":
false, "item": "cinder-backup/ceph.client.cinder.keyring", "msg": "Could not find or access
/etc/kolla/config/cinder/cinder-backup/ceph.client.cinder.keyring" on the Ansible Controller.\nIf you are using a
module and expect the file to exist on the remote, see the remote_src option"}
An exception occurred during task execution. To see the full traceback, use -vvv. The error was: If you are using a
module and expect the file to exist on the remote, see the remote_src option
failed: [GPU111] (item=cinder-backup/ceph.client.cinder-backup.keyring) => {"ansible_loop_var": "item",
"changed": false, "item": "cinder-backup/ceph.client.cinder-backup.keyring", "msg": "Could not find or access
/etc/kolla/config/cinder/cinder-backup/ceph.client.cinder-backup.keyring" on the Ansible Controller.\nIf you are
using a module and expect the file to exist on the remote, see the remote_src option"}
```

Setup Ceph client authentication to create cinder user for accessing volumes and vms pool.

```
root@GPU111:~# ceph auth get-or-create client.cinder mon 'profile rbd' osd 'profile rbd pool=volumes, profile rbd pool=vms' mgr
'profile rbd pool=volumes, profile rbd pool=vms'
[client.cinder]

key = AQAC5nBjU4CrCRAAi3kUAhlpOz5dvWZKg9pNw==
```

Also for cinder-backup. (We still use NFS for cinder-backup anyway)

```
root@GPU111:/etc/kolla# ceph auth get-or-create client.cinder-backup mon 'profile rbd' osd 'profile rbd pool=backups' mgr 'profile rbd
pool=backups'
[client.cinder-backup]

key = AQB++HBjbpHIBAAi68wc04/vfhYJGmNgTQoQ==
```

Copy the key to deploy node for cinder

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ mkdir -p ./config/cinder/cinder-volume/

(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cat ./config/cinder/cinder-volume/ceph.client.cinder.keyring
[client.cinder]

key = AQAC5nBjU4CrCRAAi3kUAhlpOz5dvWZKg9pNw==
```


Deploy OpenStack using Kolla-Ansible (Cons't)

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ mkdir -p ./config/cinder/cinder-backup/
```

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cp ./config/cinder/cinder-volume/ceph.client.cinder.keyring  
./config/cinder/cinder-backup/
```

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cat ./config/cinder/cinder-backup/ceph.client.cinder-backup.keyring  
[client.cinder-backup]
```

```
key = AQB++HBjprHIBAAi68wc04/vfhYJGmNgTQoQ==
```

NOTE: cinder-backup needs cinder and cinder-backup keys.

Copy ceph.conf to deploy node

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cat ./config/cinder/ceph.conf  
# minimal ceph.conf for 16e21856-6030-11ed-8a24-db02e0e54b22  
[global]  
fsid = 16e21856-6030-11ed-8a24-db02e0e54b22  
mon_host = [v2:192.168.1.23:3300/0,v1:192.168.1.23:6789/0]
```

NOTE: remove the "tab" in the ceph.conf. Otherwise kolla-ansible would fail with error.

```
TASK [cinder : Copying over ceph.conf for Cinder]
```

```
An exception occurred during task execution. To see the full traceback, use -vvv. The error was:  
oslo_config.iniparser.ParseError: at line 3, Unexpected continuation line: '\tfsid = 16e21856-6030-11ed-8a24-db02e0e54b22'  
fatal: [GPU111]: FAILED! => {"msg": "Unexpected failure during module execution.", "stdout": ""}
```

```
[global]  
fsid =  
mon_host =
```

Copy ceph.conf and cinder keyring for Nova

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ mkdir -p ./config/nova
```

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cp ./config/cinder/ceph.conf ./config/nova/  
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ cp ./config/cinder/cinder-volume/ceph.client.cinder.keyring ./config/nova/
```

Check Ceph config files for kolla

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ tree ./config/  
./config/  
├── cinder  
│   ├── ceph.conf  
│   ├── cinder-backup  
│   │   ├── ceph.client.cinder-backup.keyring  
│   │   └── ceph.client.cinder.keyring  
│   └── cinder-volume  
│       └── ceph.client.cinder.keyring  
├── nfs_shares  
└── nova  
    ├── ceph.client.cinder.keyring  
    └── ceph.conf
```

Config globals.yml

```
cinder_backend_ceph: "yes"  
nova_backend_ceph: "yes"
```

```
ceph_cinder_keyring: "ceph.client.cinder.keyring"  
ceph_cinder_user: "cinder"  
ceph_cinder_pool_name: "volumes"  
ceph_cinder_backup_keyring: "ceph.client.cinder-backup.keyring"  
ceph_cinder_backup_user: "cinder-backup"  
ceph_cinder_backup_pool_name: "backups"
```

```
ceph_nova_keyring: "{{ ceph_cinder_keyring }}"  
ceph_nova_user: "{{ ceph_cinder_user }}"  
ceph_nova_pool_name: "vms"
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Apply configuration

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-ansible -i multinode deploy
```

Check cinder volume service status

```
root@GPU111:~# openstack volume service list
+-----+-----+-----+-----+-----+-----+
| Binary | Host | Zone | Status | State | Updated At |
+-----+-----+-----+-----+-----+-----+
| cinder-scheduler | GPU111 | nova | enabled | up | 2025-18-13T14:36:43.000000 |
| cinder-volume | GPU111@nfs-1 | nova | enabled | up | 2025-18-13T14:36:44.000000 |
| cinder-backup | GPU111 | nova | enabled | up | 2025-18-13T14:36:48.000000 |
| cinder-volume | GPU111@rbd-1 | nova | enabled | up | 2025-18-13T14:36:47.000000 |
+-----+-----+-----+-----+-----+-----+
```

The cinder-backup use NFS backup driver

```
root@GPU111:/etc/kolla/cinder-backup# cat cinder.conf | grep backup
backup_driver = cinder.backup.drivers.nfs.NFSBackupDriver
backup_mount_options =
backup_mount_point_base = /var/lib/cinder/backup
backup_share = GPU111:/kolla_nfs
backup_file_size = 327680000
```

Cinder volume has two backends enabled

```
root@GPU111:/etc/kolla/cinder-volume# cat cinder.conf | grep enabled_backends
enabled_backends = rbd-1,nfs-1
```

```
[rbd-1]
volume_driver = cinder.volume.drivers.rbd.RBDDriver
volume_backend_name = rbd-1
rbd_pool = volumes
rbd_ceph_conf = /etc/ceph/ceph.conf
rados_connect_timeout = 5
rbd_user = cinder
rbd_secret_uuid = a8c5fb21-a2da-46a3-9275-2c3365bc3a12
report_discard_supported = True

[nfs-1]
volume_driver = cinder.volume.drivers.nfs.NfsDriver
volume_backend_name = nfs-1
nfs_shares_config = /etc/cinder/nfs_shares
nfs_snapshot_support = True
nas_secure_file_permissions = False
nas_secure_file_operations = False
```

Create volume types

```
root@GPU111:/etc/kolla/cinder-volume# openstack volume type create ceph
+-----+-----+-----+-----+-----+-----+
| Field | Value |
+-----+-----+-----+-----+-----+-----+
| description | None |
| id | f2c1ce62-d33f-4045-b94b-fce501961679 |
| is_public | True |
| name | ceph |
+-----+-----+-----+-----+-----+-----+

root@GPU111:/etc/kolla/cinder-volume# openstack volume type set ceph --property volume_backend_name=rbd-1
```

Deploy OpenStack using Kolla-Ansible (Cons't)

```
root@GPU111:/etc/kolla/cinder-volume# openstack volume type list --long
```

ID	Name	Is Public	Description	Properties
f2c1ce62-d33f-4045-b94b-fce501961679	ceph	True	None	volume_backend_name='rbd-1'
f9782ab2-931d-430b-92dd-b6810677c0e0	__DEFAULT__	True	Default Volume Type	

Test VM creation with ceph volume type

```
root@GPU111:/etc/kolla/cinder-volume# openstack flavor list
```

ID	Name	RAM	Disk	Ephemeral	VCPUs	Is Public
1	m1.tiny	512	1	0	1	True
2	m1.small	2048	20	0	1	True
3	m1.medium	4096	40	0	2	True
4	m1.large	8192	80	0	4	True
5	m1.xlarge	16384	160	0	8	True
8f99ab4a-e2de-493d-9940-37de21df623	1c2r4d	2048	4	0	1	False

```
root@GPU111:/etc/kolla/cinder-volume# openstack keypair list
```

Name	Fingerprint
key-pair1	31:a1:03:89:15:fb:5b:46:e7:b8:5a:94:8d:39:82:12

```
root@GPU111:/etc/kolla/cinder-volume# openstack image list
```

ID	Name	Status
1fa43af3-ac22-4dba-85e9-04b21807c5d4	cirros	active
40cbae5f-79ab-4c4f-ba97-fc97c6cd65f4	ubuntu22.04	active

```
root@GPU111:/etc/kolla/cinder-volume# openstack network list
```

ID	Name	Subnets
17cb537a-88ac-42ed-9eb2-29177cb7490e	public1	142f17ca-c06d-4c07-abdc-dc935b1b83a3
67e89302-e5c7-4a17-ad1c-675b664874c1	demo-net	312eff74-d80c-4281-a51a-9b4bed5132c4

Create a VM with "ephemeral" disk.

```
root@GPU111:~# openstack server create --flavor 1c2r4d --image ubuntu22.04 --key-name key-pair1 --network demo-net ubuntu_vm
```

Field	Value
OS-DCF:diskConfig	MANUAL
OS-EXT-AZ:availability_zone	
OS-EXT-SRV-ATTR:host	None
OS-EXT-SRV-ATTR:hypervisor_hostname	None
OS-EXT-SRV-ATTR:instance_name	
OS-EXT-STS:power_state	NOSTATE
OS-EXT-STS:task_state	scheduling
OS-EXT-STS:vm_state	building
OS-SRV-USG:launched_at	None
OS-SRV-USG:terminated_at	None
accessIPv4	
accessIPv6	
addresses	
adminPass	s5vdJaDqsBh7
config_drive	
created	2025-18-14T11:58:01Z
flavor	1c2r4d (8f99ab4a-e2de-493d-9940-37de21df623)
hostid	
id	1adcbbf1f-0dd9-4086-a67f-ccf73d8b94e6
image	ubuntu22.04 (40cbae5f-79ab-4c4f-ba97-fc97c6cd65f4)
key_name	key-pair1
name	ubuntu_vm
progress	0
project_id	496d5b465e684f20a8217fa408728ac2
properties	
security_groups	name='default'
status	BUILD
updated	2025-18-14T11:58:01Z
user_id	6714a667d1e3406fab0c7add7d71d412
volumes_attached	

```
root@GPU111:~# openstack server show ubuntu_vm
```

Field	Value
OS-DCF:diskConfig	MANUAL
OS-EXT-AZ:availability_zone	nova
OS-EXT-SRV-ATTR:host	GPU72
OS-EXT-SRV-ATTR:hypervisor_hostname	GPU7CJYHN2
OS-EXT-SRV-ATTR:instance_name	instance-000000007
OS-EXT-STS:power_state	Running
OS-EXT-STS:task_state	None
OS-EXT-STS:vm_state	active

Deploy OpenStack using Kolla-Ansible (Cons't)

Test live migration

```
root@GPU111:~# openstack server migrate --live-migration ubuntu_vm
root@GPU111:~# openstack server show ubuntu_vm
```

Field	Value
OS-DCF:diskConfig	MANUAL
OS-EXT-AZ:availability_zone	nova
OS-EXT-SRV-ATTR:host	GPU71
OS-EXT-SRV-ATTR:hypervisor_hostname	GPU7CJYHN1
OS-EXT-SRV-ATTR:instance_name	instance-000000007
OS-EXT-STS:power_state	Running
OS-EXT-STS:task_state	None
OS-EXT-STS:vm_state	active
OS-SRV-USG:launched_at	2025-18-14T11:59:04.000000
OS-SRV-USG:terminated_at	None
accessIPv4	
accessIPv6	
addresses	demo-net=10.0.0.142
config_drive	
created	2025-18-14T11:58:01Z
flavor	1c2r4d (8f99ab4a-e2de-493d-9940-37de21dfd623)
hostid	c7db1b2553c921575d60036b5d792993c9a242db70405cb52bc0bfea
id	1adcbf1f-0dd9-4086-a67f-ccf73d8b94e6
image	ubuntu22.04 (40cbae5f-79ab-4c4f-ba97-fc97c6cd65f4)
key_name	key-pair1
name	ubuntu_vm
progress	0
project_id	496d5b465e684f20a8217fa408728ac2
properties	
security_groups	name='default'
status	ACTIVE
updated	2025-18-14T12:33:46Z
user_id	6714a667d1e3406fab0c7add7d71d412
volumes_attached	

Create cinder volume on ceph

```
root@GPU111:~# openstack volume create --bootable --image ubuntu22.04 --type ceph --size 3 ubuntu22.04_boot_image
```

Field	Value
attachments	[]
availability_zone	nova
bootable	false
consistencygroup_id	None
created_at	2025-18-14T12:37:31.146975
description	None
encrypted	False
id	6c4f5912-4480-46cd-a960-f0a59a3f5d98
migration_status	None
multiattach	False
name	ubuntu22.04_boot_image
properties	
replication_status	None
size	3
snapshot_id	None
source_vol_id	None
status	creating
type	ceph
updated_at	None
user_id	6714a667d1e3406fab0c7add7d71d412

```
root@GPU111:~# openstack volume list --long
```

ID	Name	Status	Size	Type	Bootable	Attached to	Properties
6c4f5912-4480-46cd-a960-f0a59a3f5d98	ubuntu22.04_boot_image	available	3	ceph	true		

```
root@GPU111:~# rbd ls -l volumes
```

NAME	SIZE	PARENT	FMT	PROT	LOCK
volume-6c4f5912-4480-46cd-a960-f0a59a3f5d98	3 GiB				2

Deploy OpenStack using Kolla-Ansible (Cons't)

Create VM with the volume

```
root@GPU111:~# openstack server create --volume ubuntu22.04_boot_image --flavor 1c2r4d --key-name key-pair1 --network demo-net ubuntu22.04_ceph_volume_vm
```

Field	Value
OS-DCF:diskConfig	MANUAL
OS-EXT-AZ:availability_zone	
OS-EXT-SRV-ATTR:host	None
OS-EXT-SRV-ATTR:hypervisor_hostname	None
OS-EXT-SRV-ATTR:instance_name	
OS-EXT-STS:power_state	NOSTATE
OS-EXT-STS:task_state	scheduling
OS-EXT-STS:vm_state	building
OS-SRV-USG:launched_at	None
OS-SRV-USG:terminated_at	None
accessIPv4	
accessIPv6	
addresses	
adminPass	vTNp7WWrhuFi
config_drive	
created	2025-18-14T12:41:40Z
flavor	1c2r4d (8f99ab4a-e2de-493d-9940-37de21dfd623)
hostId	
id	b7143bbd-f23e-42f9-9125-0b61592602f0
image	
key_name	key-pair1
name	ubuntu22.04_ceph_volume_vm
progress	0
project_id	496d5b465e684f20a8217fa408728ac2
properties	
security_groups	name='default'
status	BUILD
updated	2025-18-14T12:41:40Z
user_id	6714a667d1e3406fab0c7add7d71d412
volumes_attached	

```
root@GPU111:~# openstack server list
```

ID	Name	Status	Networks	Image	Flavor
b7143bbd-f23e-42f9-9125-0b61592602f0	ubuntu22.04_ceph_volume_vm	ACTIVE	demo-net=10.0.0.182		1c2r4d
1adcbf1f-0dd9-4086-a67f-ccf73d8b94e6	ubuntu_vm	ACTIVE	demo-net=10.0.0.142	ubuntu22.04	1c2r4d

Extra: Use local docker registry to deploy (Not Work)

Create a local docker registry on host GPU10 for Kolla-Ansible

NOTE: This does not work because kolla-ansible use "quay.io" to pull images and under "openstack.kolla" organization. However the docker registry <https://registry-1.docker.io> does not have "openstack.kolla". This results image pull failure.

Error message examples:

```
root@GPU111:~# docker pull GPU10:5000/openstack.kolla/fluentd:master-rocky-9
Error response from daemon: manifest for GPU10:5000/openstack.kolla/fluentd:master-rocky-9 not found: manifest unknown: manifest unknown
```

```
docker.errors.NotFound: 404 Client Error for http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A5000%2Fopenstack.kolla%2Ffluentd: Not Found (\"manifest for GPU10:5000/openstack.kolla/fluentd:master-rocky-9 not found: manifest unknown: manifest unknown)
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Create /etc/registry directory

```
root@GPU10:~# mkdir -p /etc/docker/registry
root@GPU10:~# cd /etc/docker/registry
root@GPU10:/etc/docker/registry#
```

Create docker-compose.yml file

```
root@GPU10:/etc/docker/registry# cat docker-compose.yml
version: '3'

services:
  registry:
    container_name: registry
    image: registry:2
    ports:
      - "5000:5000"
    environment:
      REGISTRY_STORAGE_FILESYSTEM_ROOTDIRECTORY: /data
    volumes:
      - /mnt/docker-registry-data:/data
      - /etc/docker/registry/config.yml:/etc/docker/registry/config.yml
```

Create config.yml file

```
root@GPU10:/etc/docker/registry# cat ./config.yml
version: 0.1
log:
  fields:
    service: registry
storage:
  cache:
    blobdescriptor: inmemory
  filesystem:
    rootdirectory: /var/lib/registry
http:
  addr: :5000
  headers:
    X-Content-Type-Options: [nosniff]
health:
  storagedriver:
    enabled: true
    interval: 10s
    threshold: 3
proxy:
  remoteurl: https://registry-1.docker.io
```

Run docker registry container

```
root@GPU10:/etc/docker/registry# docker-compose up -d
Creating network "registry_default" with the default driver
Pulling registry (registry:2)...
2: Pulling from library/registry
ca7dd9ec2225: Pull complete
c41ae7ad2b39: Pull complete
1ed0fc8a6161: Pull complete
21df229223d2: Pull complete
626897ccab21: Pull complete
Digest: sha256:ce14a6258f37702ff3cd92232a6f5b81ace542d9f1631966999e9f7c1ee6ddba
Status: Downloaded newer image for registry:2
Creating registry ... done
```

Configure insecure-registries and registry-mirrors

```
root@GPU10:/etc/docker/registry# cat /etc/docker/daemon.json
{
  "insecure-registries": ["GPU10:5000"],
  "registry-mirrors": ["http://GPU10:5000"]
}
```

```
root@GPU10:/etc/docker/registry# systemctl restart docker
```

Test docker registry

```
root@GPU10:/etc/docker/registry# docker pull fedora
Using default tag: latest
latest: Pulling from library/fedora
cb8b1ed77979: Pull complete
Digest: sha256:f99efcdcd4dd6736d8a88cc1ab6722098ec1d77dbf7aed9a7a514fc997ca08e0
Status: Downloaded newer image for fedora:latest
docker.io/library/fedora:latest
```

```
root@GPU10:/etc/docker/registry# du -sh /mnt/docker-registry-data/docker/registry/v2/
64M      /mnt/docker-registry-data/docker/registry/v2/
```

NOTE: docker will query configured registry mirror which trigger mirror download image. You should see image data usage increased in `"/mnt/docker-registry-data/docker/registry/v2/blobs/"`)

Deploy OpenStack using Kolla-Ansible (Cons't)

If test success, configure `/etc/docker/daemon.json` on all OpenStack nodes. For example on controller host:

```
ubuntu@GPU111:~$ cat /etc/docker/daemon.json
{
  "bridge": "none",
  "ip-forward": false,
  "iptables": false,
  "insecure-registries": ["GPU10:5000"],
  "registry-mirrors": ["http://GPU10:5000"]
}

ubuntu@GPU111:~$ sudo systemctl restart docker
```

Edit globals.yml

```
docker_registry: GPU10:5000
docker_registry_insecure: "yes"
```

Run kolla-ansible

```
(kolla-ansible-venv) amel@GPU10:/etc/kolla$ kolla-ansible -i multinode deploy
```

NOTE: Do not run kolla-ansible precheck, it removes "registry-mirrors" in docker daemon config.

NOTE: We have to set "insecure-registries" because we do not use https.

Troubleshoot

Error when use local docker registry.

```
RUNNING HANDLER [common : Restart fluentd container]
*****
fatal: [GPU111]: FAILED! => {"changed": true, "msg": "'Traceback (most recent call last):\\n File \\\"/usr/local/lib/python3.8/dist-packages/docker/api/client.py\\\", line 268, in _raise_for_status\\n   response.raise_for_status()\\n File \\\"/usr/lib/python3/dist-packages/requests/models.py\\\", line 940, in raise_for_status\\n   raise HTTPError(http_error_msg, response=self)\\nrequests.exceptions.HTTPError: 500 Server Error: Internal Server Error for url: http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A4000%2Fopenstack.kolla%2Ffluentd\\n\\nDuring handling of the above exception, another exception occurred:\\n\\nTraceback (most recent call last):\\n File \\\"/tmp/ansible_kolla_docker_payload_hl3f6zvw/ansible_kolla_docker_payload.zip/ansible/modules/kolla_docker.py\\\", line 381, in main\\n File \\\"/tmp/ansible_kolla_docker_payload_hl3f6zvw/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\\\", line 660, in recreate_or_restart_container\\n   self.pull_image()\\n File \\\"/tmp/ansible_kolla_docker_payload_hl3f6zvw/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\\\", line 450, in pull_image\\n   json.loads(line.strip()).decode(\\\"utf-8\\\") for line in self.dc.pull(\\n File \\\"/usr/local/lib/python3.8/dist-packages/docker/api/image.py\\\", line 430, in pull\\n   self._raise_for_status(response)\\n File \\\"/usr/local/lib/python3.8/dist-packages/docker/api/client.py\\\", line 270, in _raise_for_status\\n   raise create_api_error_from_http_exception(e)\\n File \\\"/usr/local/lib/python3.8/dist-packages/docker/errors.py\\\", line 31, in create_api_error_from_http_exception\\n   raise cls(e, response=response, explanation=explanation)\\n docker.errors.APIError: 500 Server Error for http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A4000%2Fopenstack.kolla%2Ffluentd: Internal Server Error (\\\"Get \\\"https://GPU10:4000/v2/\\\": http: server gave HTTP response to HTTPS client\\\")\\n\"}
```

Deploy OpenStack using Kolla-Ansible (Cons't)

Troubleshoot

Error when use local docker registry.

```
fatal: [GPU72]: FAILED! => {"changed": true, "msg": "'Traceback (most recent call last):\\n File \\usr/local/lib/python3.8/dist-packages/docker/api/client.py\\n, line 268, in _raise_for_status\\n response.raise_for_status()\\n File \\usr/lib/python3/dist-packages/requests/models.py\\n, line 940, in raise_for_status\\n raise HTTPError(http_error_msg, response=self)\\nrequests.exceptions.HTTPError: 500 Server Error: Internal Server Error for url: http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A4000%2Fopenstack.kolla%2Ffluentd\\n\\nDuring handling of the above exception, another exception occurred:\\n\\nTraceback (most recent call last):\\n File \\tmp/ansible_kolla_docker_payload_h4_q1cuz/ansible_kolla_docker_payload.zip/ansible/modules/kolla_docker.py\\n, line 381, in main\\n File \\tmp/ansible_kolla_docker_payload_h4_q1cuz/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\\n, line 660, in recreate_or_restart_container\\n self.pull_image()\\n File \\tmp/ansible_kolla_docker_payload_h4_q1cuz/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\\n, line 450, in pull_image\\n json.loads(line.strip().decode(\\utf-8\\')) for line in self.dc.pull(\\n File \\usr/local/lib/python3.8/dist-packages/docker/api/image.py\\n, line 430, in pull\\n self._raise_for_status(response)\\n File \\usr/local/lib/python3.8/dist-packages/docker/api/client.py\\n, line 270, in _raise_for_status\\n raise create_api_error_from_http_exception(e)\\n File \\usr/local/lib/python3.8/dist-packages/docker/errors.py\\n, line 31, in create_api_error_from_http_exception\\n raise cls(e, response=response, explanation=explanation)\\ndocker.errors.APIError: 500 Server Error for http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A4000%2Fopenstack.kolla%2Ffluentd: Internal Server Error (\\\"Get \\\"https://GPU10:4000/v2/\\\": x509: certificate relies on legacy Common Name field, use SANs instead\\\")\\n\"}
```

```
fatal: [GPU111]: FAILED! => {"changed": true, "msg": "'Traceback (most recent call last):\\n File \\usr/local/lib/python3.8/dist-packages/docker/api/client.py\\n, line 268, in _raise_for_status\\n response.raise_for_status()\\n File \\usr/lib/python3/dist-packages/requests/models.py\\n, line 940, in raise_for_status\\n raise HTTPError(http_error_msg, response=self)\\nrequests.exceptions.HTTPError: 404 Client Error: Not Found for url: http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A5000%2Fopenstack.kolla%2Ffluentd\\n\\nDuring handling of the above exception, another exception occurred:\\n\\nTraceback (most recent call last):\\n File \\tmp/ansible_kolla_docker_payload_s36ljb33/ansible_kolla_docker_payload.zip/ansible/modules/kolla_docker.py\\n, line 381, in main\\n File \\tmp/ansible_kolla_docker_payload_s36ljb33/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\\n, line 651, in recreate_or_restart_container\\n self.start_container()\\n File \\tmp/ansible_kolla_docker_payload_s36ljb33/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\\n, line 669, in start_container\\n self.pull_image()\\n File \\tmp/ansible_kolla_docker_payload_s36ljb33/ansible_kolla_docker_payload.zip/ansible/module_utils/kolla_docker_worker.py\\n, line 450, in pull_image\\n json.loads(line.strip().decode(\\utf-8\\')) for line in self.dc.pull(\\n File \\usr/local/lib/python3.8/dist-packages/docker/api/image.py\\n, line 430, in pull\\n self._raise_for_status(response)\\n File \\usr/local/lib/python3.8/dist-packages/docker/api/client.py\\n, line 270, in _raise_for_status\\n raise create_api_error_from_http_exception(e)\\n File \\usr/local/lib/python3.8/dist-packages/docker/errors.py\\n, line 31, in create_api_error_from_http_exception\\n raise cls(e, response=response, explanation=explanation)\\ndocker.errors.NotFound: 404 Client Error for http+docker://localhost/v1.41/images/create?tag=master-rocky-9&fromImage=GPU10%3A5000%2Fopenstack.kolla%2Ffluentd: Not Found (\\\"manifest for GPU10:5000/openstack.kolla/fluentd:master-rocky-9 not found: manifest unknown: manifest unknown\\\")\\n\"}
```

Troubleshoot

Error when use local docker registry.

- kolla-ansible -i INVENTORY deploy is used to deploy and start all Kolla containers.
- kolla-ansible -i INVENTORY destroy is used to clean up containers and volumes in the cluster.
- kolla-ansible -i INVENTORY mariadb_recovery is used to recover a completely stopped mariadb cluster.
- kolla-ansible -i INVENTORY prechecks is used to check if all requirements are met before deploy for each of the OpenStack services.
- kolla-ansible -i INVENTORY post-deploy is used to do post deploy on deploy node to get the admin openrc file.
- kolla-ansible -i INVENTORY pull is used to pull all images for containers.
- kolla-ansible -i INVENTORY reconfigure is used to reconfigure OpenStack service.
- kolla-ansible -i INVENTORY upgrade is used to upgrade existing OpenStack Environment.
- kolla-ansible -i INVENTORY stop is used to stop running containers.
- kolla-ansible -i INVENTORY deploy-containers is used to check and if necessary update containers, without generating configuration.
- kolla-ansible -i INVENTORY prune-images is used to prune orphaned Docker images on hosts.
- kolla-ansible -i INVENTORY genconfig is used to generate configuration files for enabled OpenStack services, without then restarting the containers so it is not applied right away.
- kolla-ansible -i INVENTORY1 -i INVENTORY2 ... Multiple inventories can be specified by passing the --inventory or -i command line option multiple times. This can be useful to share configuration between multiple environments. Any common configuration can be set in INVENTORY1 and INVENTORY2 can be used to set environment specific details.
- kolla-ansible -i INVENTORY gather-facts is used to gather Ansible facts, for example to populate a fact cache.

Mauliate Godang